Environmental Research 241 (2024) 117630

Contents lists available at ScienceDirect

Environmental Research

journal homepage: www.elsevier.com/locate/envres

Review article

Insights into current bio-processes and future perspectives of carbon-neutral treatment of industrial organic wastewater: A critical review

Xinzheng Zhang [a,b], Yuqin Fan [a,b], Tianwei Hao [c], Rong Chen [d], Tao Zhang [e], Yong Hu [f], Dapeng Li [a,b], Yang Pan [a,b], Yu-You Li [g], Zhe Kong [a,b,*]

[a] National and Local Joint Engineering Laboratory of Municipal Sewage Resource Utilization Technology, School of Environmental Science and Engineering, Suzhou University of Science and Technology, Suzhou, 215009, China
[b] Jiangsu Collaborative Innovation Center of Water Treatment Technology and Material, School of Environmental Science and Engineering, Suzhou University of Science and Technology, Suzhou, 215009, China
[c] Department of Civil and Environmental Engineering, Faculty of Science and Technology, University of Macau, Macau, China
[d] Key Lab of Environmental Engineering, Shaanxi Province, Xi'an University of Architecture and Technology, Xi'an, 710055, China
[e] College of Design and Innovation, Shanghai International College of Design & Innovation, Tongji University, Shanghai, 200092, China
[f] School of Environmental Science and Engineering, Nanjing Tech University, Nanjing, 211816, China
[g] Department of Civil and Environmental Engineering, Graduate School of Engineering, Tohoku University, 6-6-06 Aza-Aoba, Aramaki, Aoba Ward, Sendai, Miyagi, 980-8579, Japan

ARTICLE INFO

Keywords:
Industrial organic wastewater
Carbon neutrality
Anaerobic digestion
Bioenergy recovery
AnMBR
Anammox

ABSTRACT

With the rise of the concept of carbon neutrality, the current wastewater treatment process of industrial organic wastewater is moving towards the goal of energy conservation and carbon emission reduction. The advantages of anaerobic digestion (AD) processes in industrial organic wastewater treatment for bio-energy recovery, which is in line with the concept of carbon neutrality. This study summarized the significance and advantages of the state-of-the-art AD processes were reviewed in detail. The application of expanded granular sludge bed (EGSB) reactors and anaerobic membrane bioreactor (AnMBR) were particularly introduced for the effective treatment of industrial organic wastewater treatment due to its remarkable prospect of engineering application for the high-strength wastewater. This study also looks forward to the optimization of the AD processes through the enhancement strategies of micro-aeration pretreatment, acidic-alkaline pretreatment, co-digestion, and biochar addition to improve the stability of the AD system and energy recovery from of industrial organic wastewater. The integration of anaerobic ammonia oxidation (Anammox) with the AD processes for the post-treatment of nitrogenous pollutants for the industrial organic wastewater is also introduced as a feasible carbon-neutral process. The combination of AnMBR and Anammox is highly recommended as a promising carbon-neutral process for the removal of both organic and inorganic pollutants from the industrial organic wastewater for future perspective. It is also suggested that the AD processes combined with biological hydrogen production, microalgae culture, bioelectrochemical technology and other bio-processes are suitable for the low-carbon treatment of industrial organic wastewater with the concept of carbon neutrality in future.

1. Introduction

In recent years, with the rise of the concept of carbon neutrality, the commitment to achieve a reduction in carbon emissions and energy consumption has also become the goals in the treatment of wastewater (L. J. Li et al., 2022; Sun et al., 2022). Due to the rapid growth of modern industrialization, the amount of industrial organic wastewater discharged is increasing every year (Mohd Udaiyappan et al., 2017; Singh et al., 2022). In 2019, global industry accounted for 16% of the world's total water consumption and discharged approximately 630×10^9 m^3 of industrial organic wastewater (Toczyłowska-Mamińska and Mamiński, 2022). The main sources of industrial organic wastewater include the food processing, paper making, textile, chemical, pharmaceutical, petroleum, leather, metallurgy and other industries (Nidheesh et al., 2020). As show in Table 1, industrial organic wastewater is typically characterized by large quantities and high concentrations of various

* Corresponding author. School of Environmental Science and Engineering, Suzhou University of Science and Technology, Suzhou, 215009, China.
E-mail address: zhekong@mail.usts.edu.cn (Z. Kong).

https://doi.org/10.1016/j.envres.2023.117630
Received 27 August 2023; Received in revised form 5 November 2023; Accepted 7 November 2023
Available online 20 November 2023
0013-9351/© 2023 Elsevier Inc. All rights reserved.

organic pollutants (defined as COD) and inorganic pollutants (typically in the form of ammonium) (S. L. Wang et al., 2022). Fig. 1 presents the treatment facilities, daily treatment capacity, and total COD and ammonium emissions of industrial organic wastewater from different sources in China in 2021 (NBS and MEE, 2023). Wastewater treatment is an energy-intensive process that requires significant energy expenditure, especially for the treatment of high-strength industrial organic wastewater (Yang et al., 2020). Energy consumption in wastewater treatment processes is reported to account for 25% of the total power consumption in the water sector worldwide, even 42% in most developed countries, and is expected to reach more than 60% by 2040 (di Cicco et al., 2019). This proportion of energy is typically sourced from electricity generated by burning fossil fuels, which releases large amounts of greenhouse gases (Żyłka et al., 2021). However, if wastewater can be viewed as a potentially sustainable energy source, it contains at least two to four times the energy required to treat the same wastewater (Koul et al., 2022). Therefore, it would be significantly beneficial if the energy contained in wastewater can be fully utilized to meet the energy demand of wastewater (Yadav et al., 2021). As a result, the principle of "carbon neutrality" in the wastewater treatment processes can be realized in the forms of zero greenhouse gas emissions, energy self-sufficiency and even extra production of sustainable bio-energy throughout the entire life cycle of wastewater treatment process (Hao et al., 2015).

While industrial organic wastewater can be effectively treated by state-of-the-art chemical methods such as advanced oxidation process (AOP) (Saravanan et al., 2022), sustainability should be taken into. With the advantages of rapid reaction, complete degradation and high efficiency, AOPs such as the Fenton process have been widely used to treat various types of industrial organic wastewater (Cheng et al., 2022), especially those containing refractory and persistent organic pollutants. However, the use of chemicals and oxidants in massive treatment of industrial organic wastewater containing high concentrations of organic pollutants can be highly expensive. Besides, when treating high strength industrial organic wastewater, strong oxidants degrade the organic matter directly to CO_2 and do not achieve carbon reduction.

Biological processes are considered to be a more suitable method for the treatment of industrial organic wastewater (Meena et al., 2022). While the conventional activated sludge (CAS) method is considered to be the most widely adopted bio-process for treating all types of wastewater (Šabić Runjavec et al., 2022), it still has drawbacks such as high energy consumption, high CO_2 emission, high sludge production, and its inability to perform sustainable bio-energy recovery (Drewnowski et al., 2019; Garfí et al., 2017). Carbon neutral treatment of industrial organic wastewater should therefore be based on such low-carbon and low-cost bio-technologies and bio-processes, which have the potential of sustainability, energy conservation and carbon reduction.

In recent years, the application of anaerobic digestion (AD) processes to the treatment of high-strength industrial organic wastewater and organic solid wastes has received worldwide attention (Hassan et al., 2022; Saravanan et al., 2023). With the advantages of low energy consumption, high resistance to high organic loading rate, low sludge yield and the availability of energy neutralization through the production of biogas, the AD processes is recognized as a sustainable and low-carbon bio-technology, which is in line with the concept of carbon neutrality in wastewater treatment (Subbarao et al., 2023; Xu et al., 2021). In addition to the use of the AD processes to treat organic pollutants in industrial organic wastewater, it should also be noted that ammonium is another main pollutant in industrial organic wastewater. As industrial organic wastewater typically contains nitrogenous organic compounds such as protein, amides, antibiotics and aromatics (Guo et al., 2020; Yin et al., 2023), they are converted into ammonium after degradation by the AD process and require further post-treatment of nitrogen removal (Kuypers et al., 2018). Similar to the CAS process for the removal of organic matter, the widely used conventional nitrification/denitrification process incurs disadvantages such as high operating cost and high sludge production (Q. Zhang et al., 2022). Fortunately, a promising autotrophic bio-technology, the anaerobic ammonia oxidation (anammox) process has emerged as the most suitable solution for the carbon-neutral treatment of nitrogen-bearing industrial organic wastewater (Weralupitiya et al., 2021; Xiao et al., 2021). Anammox processes have been reported to have the ability of reducing aeration energy consumption by 60%, sludge production by 90%, greenhouse gas emissions by 90% and organic carbon source consumption by 100% (Wu et al., 2022). Therefore, to achieve carbon-neutral treatment of industrial organic wastewater, it is proposed to orient future research on the

Table 1
Properties and major pollutants of some typical industrial organic wastewater.

Wastewater Sources	pH	COD (mg/L)	BOD (mg/L)	TSS (mg/L)	TN (mg/L)	Major pollutants		Reference
						Categories	Concentration (mg/L)	
Olive mill wastewater	4.8 ± 0.2	75000 ± 290	48000 ± 140	17000 ± 500	590 ± 10	Phenols	8650 ± 30	Shabir et al. (2022)
Petrochemical wastewater	7–8.5	4600–5300	*N.A*[a]	50–150	35–40	Petrochemicals	50–80	Liang et al. (2019)
Pulp and paper wastewater	8.1 ± 0.2	25634 ± 239	8671 ± 187	89 ± 1.49	194 ± 5	Lignin	6754 ± 131	Sharma et al. (2021)
Slaughterhouse wastewater	6.7 ± 0.1	6786 ± 147	*N.A*	1700 ± 200	173 ± 1[b]	Chlortetracycline	10.236 ± 1.363[c]	Zahedi et al. (2021)
Coal chemical wastewater	*N.A*	40000–50000	*N.A*	*N.A*	3500–4500	Aromatics	14872–15139	(L. W. Yang et al., 2022)
Pharmaceutical wastewater	5.8–8.5	128–28640	20–1800	48–7500	80–164	Nalidixic Acid	45	Shah and Shah (2020)
Textile wastewater	5.5–12	100–30000	80–6000	80–10000	10–50	Dyes	*N.A*	Sharma and Malaviya (2022)
Coking wastewater	7.2–10.3	880–16800	80.6–5450	10–1460	370–1820	Phenols	25.6–1650	Tamang and Paul (2022)
Membrane Manufacturing wastewater	7.3–8.2	14000–15000	3200–4258	*N.A*	11460–12615	N, N-Dimethylacetamide	*N.A*	Xie et al. (2023)
Coal gasification wastewater	*N.A*	13000–26000	*N.A*	*N.A*	4000–12000[b]	Phenols	4000–6000	(Shi et al., 2022)

[a] Some studies did not provide the exact data.
[b] Some studies did provided information on NH_4^+-N concentrations.
[c] Derived from solid fraction in wastewater. Unit: mg/kg.

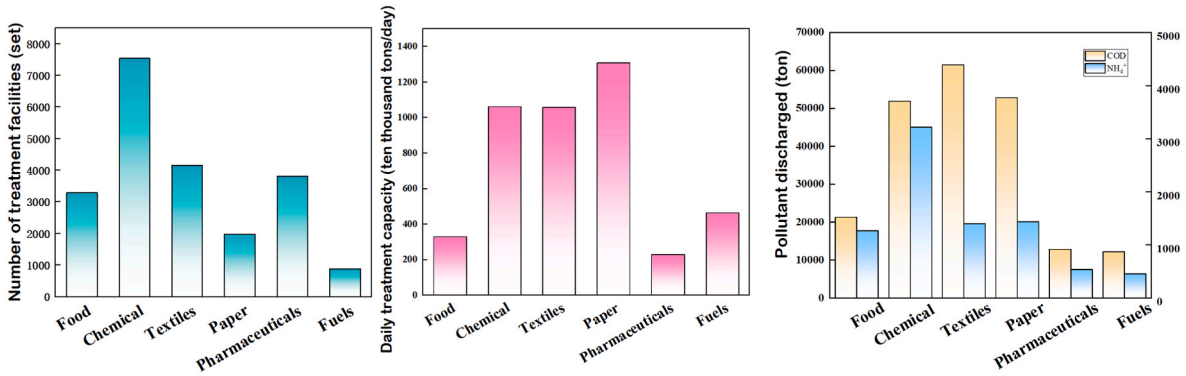


Fig. 1. (a) Number of treatment facilities in industrial organic wastewater from different sources (b) Daily treatment capacity in industrial organic wastewater from different sources treatment facilities (c) The total COD and NH_4^+ emissions contained in industrial organic wastewater from different sources in 2021.

development of bio-technology and bio-process for the removal of COD and nitrogen, based on the AD process and anammox process.

This study reviewed the current processes suitable for the carbon-neutral treatment of industrial organic wastewater, and also provides some suggested directions for the sustainable and low-carbon treatment of industrial organic wastewater with the concept of carbon neutrality, providing new insights into the future outlooks and perspectives for biological wastewater treatment. This study aimed to share a critical review from the following aspects: (1) To elucidate the difficulties of carbon-neutral treatment of industrial organic wastewater by reviewing in detail the merits and demerits of conventional treatment processes. (2) To introduce the application of the AD processes in the treatment of industrial organic wastewater and the advanced and upgraded technology and derivatives currently used to enhance the AD process under the carbon neutrality principle. (3) To highlight the importance of the post-treatment of nitrogen removal for the industrial organic wastewater and the application of promising energy-saving anammox processes, which are the most suitable solution for carbon-neutral removal of ammonia nitrogen from industrial organic wastewater. (4) To provide new insights into the future perspectives of carbon-neutral treatment of industrial organic wastewater by presenting some innovative and state-of-the-art integrated treatment technologies or strategies that could be considered as research directions for carbon-neutral treatment of industrial organic wastewater.

2. Difficulties for the conventional activated sludge process treatment of industrial organic wastewater under the concept of carbon neutrality

While the CAS process is the most commonly used bio-process for wastewater treatment, the energy consumption of the CAS process accounts for more than 45% of the total power required for a wastewater treatment plant (WWTP) (Guo et al., 2010). In most medium and large wastewater treatment plants using CAS, the energy consumption during the aeration process consumes about 45–75% of the total energy, sludge treatment requires about 15–25% of the total energy consumption, and the secondary precipitation (including circulating pumps) consumes about 15% of the total energy (Kong et al., 2021; Mamais et al., 2015). In addition, the rapid growth of aerobic microorganisms results in a large amount of excessive sludge. It has been reported that 25–50% of the influent COD is eventually converted to primary or secondary sludge, which becomes another form of waste (Leite et al., 2016). The cost of excess sludge disposal can be 25–40% of the total cost, sometimes as much as 60% (Wu et al., 2014). Table 2 presents the energy consumption of some WWTP where the CAS process is applied. It can be foreseen that for high-strength industrial organic wastewater with COD concentrations typically exceeding 10000 mg/L, the energy consumption of the CAS process is clearly unsustainable. Since electric energy is generated at the expense of burning fossil fuels, this process produces a large amount of indirect CO_2 emissions. Besides, organic pollutants are eventually converted to carbon dioxide after being treated by the CAS process, increasing direct carbon emissions (Garfí et al., 2017). The CO_2 emission of the CAS process treat of industrial organic wastewater whose COD concentration was 9648 mg/L is reported to be up to 26.64

Table 2
Energy consumption of wastewater treated by CAS process in different countries.

Countries	Energy consumption (kWh/m^3)	Contributions				Description	Reference
		Aeration (%)	Sludget reatment (%)	Pumping (%)	Other (%)		
Greece	0.128–2.280	67.2	4.3	7.5	11.6	Daily average flow is 300–27300 m^3/d, serving a population of about 1100–56000.	Siatou et al. (2020)
Japan	0.32	46	31	18	5	Daily average flow is 51263 m^3/d.	Mizuta and Shimada (2010)
Portuguese	0.23–1.30	25–80	*N.A*[a]	*N.A*	*N.A*	Avera analysis of energy consumption in 14 WWTPs in Portuguese.	Silva and Rosa (2022)
Poland	0.42–0.55	53	*N.A*	30	*N.A*	The daily average flow is 20000–25000 m^3/d, serving a population of about 200000.	Zaborowska et al. (2017)
China	0.269	*N.A*	*N.A*	*N.A*	*N.A*	Average analysis of energy consumption in 36 WWTPs in China.	Yang et al. (2010)
Italy	0.19–0.96	*N.A*	*N.A*	*N.A*	*N.A*	Survey included 241 WWTPs, serving a population of more than 9 million.	Vaccari et al. (2018)

[a] Some studies did not provide the exact data of energy consumption contributions.

kg CO_2/m^3 (Kong et al., 2022a). Therefore, the high energy consumption and high CO_2 emissions make the CAS process unsuitable as a carbon neutral technology for treating industrial organic wastewater. Although the A^2/O process, one of the upgraded derivatives of CAS, is capable of removing nitrogen from wastewater, these nitrogen-containing organic matters are usually present in the form of high concentrations of ammonium after degradation, further increasing the pressure on subsequent post-treatment processes for nitrogen removal. Therefore, more effective and sustainable bio-processes suitable to the carbon-neutral treatment of high-strength industrial organic wastewater should be of high priority.

3. Insights into the anaerobic digestion treatment of industrial organic wastewater under the concept of carbon neutrality

3.1. The significance of AD for carbon neutrality

AD is the process by which microorganisms degrade complex organic matters and produce biogas through hydrolysis, acidogenesis, acetogenesis, and methanogenesis (Y. L. Li et al., 2019). In the hydrolysis step, complex macromolecules of organic matters are first hydrolyzed by hydrolytic enzymes into easily soluble monomer. In the acidogenic step, the monomer is further decomposed into short-chain volatile fatty acids such as VFAs, lactic acid and formic acid through the function of acidogenic fermentation bacteria. In the acetogenesis step, acids with more than 2-Carbon molecule will eventually be decomposed into acetic acid and hydrogen. Finally, the acetic acid and hydrogen are converted into methane by methanogens (Ren et al., 2018).

As a carbon neutral technology, AD processes can treat industrial organic wastewater with low energy consumption due to does not require aeration and low biomass yield (Vítězová et al., 2020). Compared with the CAS process, the AD processes treat of industrial organic wastewater whose COD concentration was 3219 mg/L can reduce energy consumption by 0.71 kWh/m^3 and CO_2 emission by 0.36 kg CO_2/m^3 (Kong et al., 2022b). It is clear that the AD process can reduce more carbon emissions when treating higher concentrations of industrial organic wastewater. Another advantage of the AD process as a carbon-neutral technology is the conversion of influent COD (including toxic and refractory organic matters) into bio-methane (Kong et al., 2018). Methane is a renewable and green fuel (with a calorific value of 35.8 kJ/L) that can be obtained energy through cogeneration to offset the energy consumption required for wastewater treatment processes (R. Chen et al., 2019). Even though methane is a powerful greenhouse gas and produces large amounts of carbon emissions when burned to generate electricity, the electricity generated by burning methane significantly reduces the carbon emissions produced by burning fossil fuels (Gu et al., 2017). There are reportedly about 132,000 AD digesters (small, medium and large) worldwide, which can generate about 87 TW h of electricity every year (Karki et al., 2021). According to the aggregate carbon intensity coefficient by the year 2013 (0.52 kg CO_2/kWh) (Ang and Su, 2016). It can reduce carbon emissions by 4.524×10^{10} kg CO_2. In recent years, the application of AD to the treatment of industrial organic wastewater or wastes has gradually exhibited its significant potential in achieving carbon neutrality and energy conservation. As indicated in Table 1, industrial organic wastewater is characterized by its high COD concentration. This also implies that there is a huge potential for methane production and reduce carbon emission from the treatment of industrial organic wastewater using the AD process. It is reported that the net energy potential and relevant CO_2 emission of AD processes treat of industrial organic wastewater whose COD concentration was 9657 mg/L are 9.72 kWh/m^3 and -2.34 kg CO_2/m^3 (Kong et al., 2023). Chen et al. achieved a net energy potential of 2.12–2.82 kJ/g COD by applying the AD process to the co-digestion of coffee processing wastewater and waste activated sludge, and recycling the biogas produced (R. Chen et al., 2019). It is foreseeable that the AD process may be the most suitable solution for the carbon-neutral treatment of industrial organic wastewater in the future.

3.2. Current state-of-the-art bio-processes based on AD

3.2.1. Expanded granular sludge bed reactor

Expanded granular sludge bed (EGSB) reactor is an upgraded improved version of the up-flow anaerobic sludge blanket (UASB) process, which has all the advantages of the UASB, such as high removal efficiency, small footprint and high resistance to organic loading rate (Cruz-Salomón et al., 2020a). The EGSB process is capable of forming the anaerobic granular sludge with good settling performance through the natural coagulation and granulation inside the reactor (Faria et al., 2019), which can significantly increase the treatment capacity by reducing the distance between anaerobic microorganisms and maximizing the interspecies transfer different microbial communities (Hou et al., 2022). As shown in Fig. 2 (a), the EGSB process improves the circulation ratio and length-diameter ratio on the basis of UASB. As a result, the system can be operated at higher up-flow velocities, which avoids sludge loss and encourages the expansion of anaerobic sludge particles, and therefore significantly improves degradation efficiency (Lim and Kim, 2014). Therefore, as shown in Table 3, in recent years, the EGSB process has been considered more competitive than the UASB for the anaerobic treatment of high-strength industrial organic wastewater under high organic loading rate (OLR) conditions. Compared with the CAS process, the UASB reactor can reduce CO_2 emission by 6.18 kg/m^3 and electricity consumption by 12.20 kWh/m^3 and obtain a net energy potential of 0.061 kWh/m^3 (Kong et al., 2022a). Predictably, the EGSB reactors are capable of carrying higher organic loads, which means more methane production, and therefore greater potential for achieving carbon neutral treatment of wastewater.

However, SS is present in some industrial wastewater. Under high organic loads, SS accumulates in reactor, which affect the formation of granular sludge and weaken the performance of EGSB. The remarkable advantages of the EGSB suggest that this reactor with high bio-energy yield can be applied to treat high-strength organic wastewater that do not contain SS, such as corn alcohol wastewater, cephalosporin wastewater and cheese whey wastewater.

3.2.2. Anaerobic membrane bioreactor

Due to the slow growth rate of anaerobes, the AD process usually requires a longer HRT and SRT to ensure the growth of anaerobes and degradation of organic matter (Song et al., 2018). Membrane filtration technology retains sufficient microorganisms in the digester to avoid sludge loss and allows for separate HRT and SRT, which ultimately ensures better effluent quality (Moñino et al., 2017). Furthermore, with membrane filtration technology, no suspended matter is present in the digester, allowing for higher organic loading rates and a smaller carbon footprint (Hu et al., 2022). The net energy of the biogas produced by the AnMBR is also reported to be higher than those of the regular AD digesters (Smith et al., 2014). In a wastewater treatment plant in Japan using AnMBR, the AnMBR has shown the ability to maintain high biomass concentrations, which are 3–5 times higher than conventional AD digesters (Kanai et al., 2010). As shown in Table 3, the AnMBR process has been widely applied to the treatment of industrial organic wastewater. The significant merits of the AnMBR suggest that this carbon-neutral technology can be applied to treat high-strength organic wastewater that typically contains SS, such as meat-processing wastewater, dairy wastewater and distillery wastewater. When AnMBR treats dairy wastewater, the net energy is 51.4–53.2 kWh/m^3, and AnMBR can reduce CO_2 emission by 54.1 kg CO_2/m^3 compared to the CAS process (Ye et al., 2023). Fig. 2 (b) and **(c)** show two common AnMBR designs and three types of membrane modules.

While AnMBR is considered suitable for the treatment of high-strength industrial organic wastewater, membrane fouling caused by soluble microbial products (SMP) and extracellular polymers substances (EPS) composed of proteins and polysaccharides, which are the key

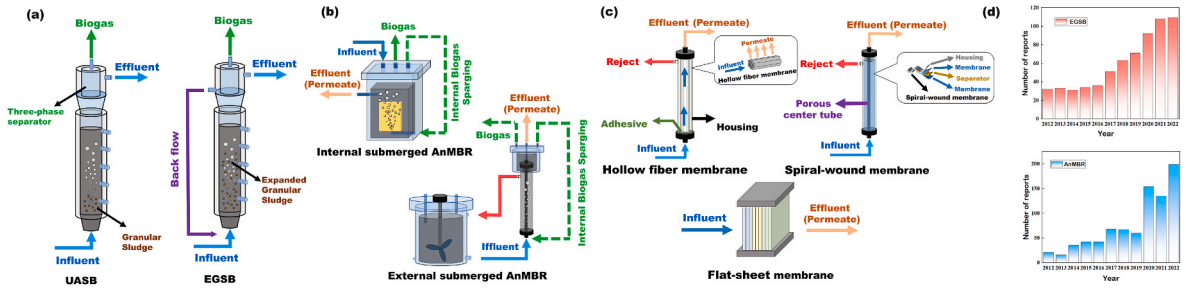


Fig. 2. (a) A conceptual graph of UASB and EGSB; (b) A conceptual graph of internal submerged AnMBR and external submerged AnMBR; (c) A conceptual graph of hollow fiber membrane, spiral-wound membrane and flat-sheet membrane; (d) Statistic numbers of reports searched in the database of Science Direct in the year 2012–2022 using the keywords "EGSB industrial wastewater" and "AnMBR industrial wastewater", respectively.

Table 3
Applications of EGSB and AnMBR process in the anaerobic treatment of industrial organic wastewater.

Wastewater Categories	Reactor type	Temperature (°C)	COD (mg/L)	OLR (kg COD/m^3/d)	HRT (days)	Removal COD (%)	CH_4 yield (L CH_4/g COD)	Reference
Cheese whey wastewater	EGSB	26.6 ± 1.4	46348 ± 3200	5.74–7.76	6–8	90–92	0.328–0.334	Cruz-Salomón et al. (2020b)
Cephalosporin wastewater	EGSB	35	5981–91870	3.60–9.96	24–25[b]	40–72	0.24–0.33	Wu et al. (2019)
Confectionery wastewater	EGSB	35	18900 ± 690	*N.A*[a]	7	89 ± 7	1000 ± 205[c]	Dereli (2019)
Corn alcohol wastewater	EGSB	35	48250–73550	14.8	1.4	95.8	10.72[d]	Zheng et al. (2022)
Tetracycline wastewater	EGSB	37	6000	5.06	24[b]	67–79	0.199	Zhang et al. (2018)
Pharmaceutical wastewater	EGSB	35 ± 2	8000–8500	*N.A*	8–20[b]	84.7–89.16	0.24–0.33	Meng et al. (2016)
Winery wastewater	AnMBR	35	5090 ± 1270	1.75 ± 0.63	2.3 ± 0.3	96.7 ± 2.7	0.33 ± 0.15	Basset et al. (2016)
Confectionery wastewater	AnMBR	35	2500–18000	1.1–7.9	*N.A*	99	0.26–0.33	Balcıoğlu et al. (2021)
Pharmaceutical wastewater	AnMBR	35	2000	3.14–6.54	8–24[b]	77.3–96.2	9.86–73.85[e]	Kong et al. (2019a)
Distillery wastewater	AnMBR	37	12200	3.97 ± 0.15	2.9 ± 0.1	95.4 ± 0.7	0.26 ± 0.01	Deschamps et al. (2021)
Malting plant wastewater	AnMBR	18–21	3238–4554	1.3–2.1	1.5–3.4	88–93	0.22–0.26[f]	Maleki (2020)
Methanolic wastewater	AnMBR	37.5 ± 1	4000–15000	5.32–20	18	88.1–90.5	1.50–7.01[d]	Lu et al. (2021)

[a] Some studies did not provide the exact data.
[b] Unit: hours.
[c] Unit: m^3/day.
[d] Volume biogas production rate. Unit: (m^3/m^3/day).
[e] Unit: mL CH_4 g/DMF g/VSS.
[f] Unit: L/g COD.

drawbacks which weaken AnMBR removal efficiency and increases operation energy consumption during the AnMBR treatment process (Anjum et al., 2021). Membrane fouling eventually results in pore blocking and subsequent cake layer formation, and is more serious and difficult to erase under the anaerobic condition (Sohn et al., 2021). The cost of membrane fouling control is a serious impediment to the potential of AnMBR technology to achieve carbon neutrality. Therefore, the search for an effective method to eliminate membrane fouling and the development of a suitable strategy for fouling control should be the key topics of future AnMBR technology research (Aslam et al., 2022).

4. Obstacles to AD for the treatment of industrial organic wastewater

4.1. Bio-toxic and degradation-resistant compounds

In general, degradable organic matters such as proteins and carbohydrates contained in industrial organic wastewater are easily decomposed into VFAs through hydrolysis and acidification processes (L. S. Wang et al., 2022).However, industrial organic wastewater from sectors such as food, textile, pharmaceutical, paper, and leather typically contains antibiotics, aromatic compounds, synthetic dyes, lignin and other toxic and refractory substances (Han et al., 2018; Sushma Kumari and Saroha, 2018). In theory, toxic and refractory organic substances can be converted into bio-energy or bio-resources through the AD process, in practice, the methane producing potential of these substances is difficult to realize due to ineffective hydrolysis. Refractory substances are usually of high molecular weight and strong chemical bonds require effective hydrolysis reaction, and the difficulty in the ineffective degradation of toxic and refractory organic compounds is the lack of corresponding hydrolytic enzymes (Kong et al., 2019b), which is shown in Fig. 3. In other words, ineffective hydrolysis of toxic and refractory substances is one of the key factors hindering treatment of industrial organic wastewater with AD process under carbon neutrality strategies. This is why a number of advanced technological tools are frequently combined with AD processes to improve the efficiency of hydrolysis.

4.2. Accumulation of intermediates and byproducts

The methane production is the decisive factor affecting the carbon

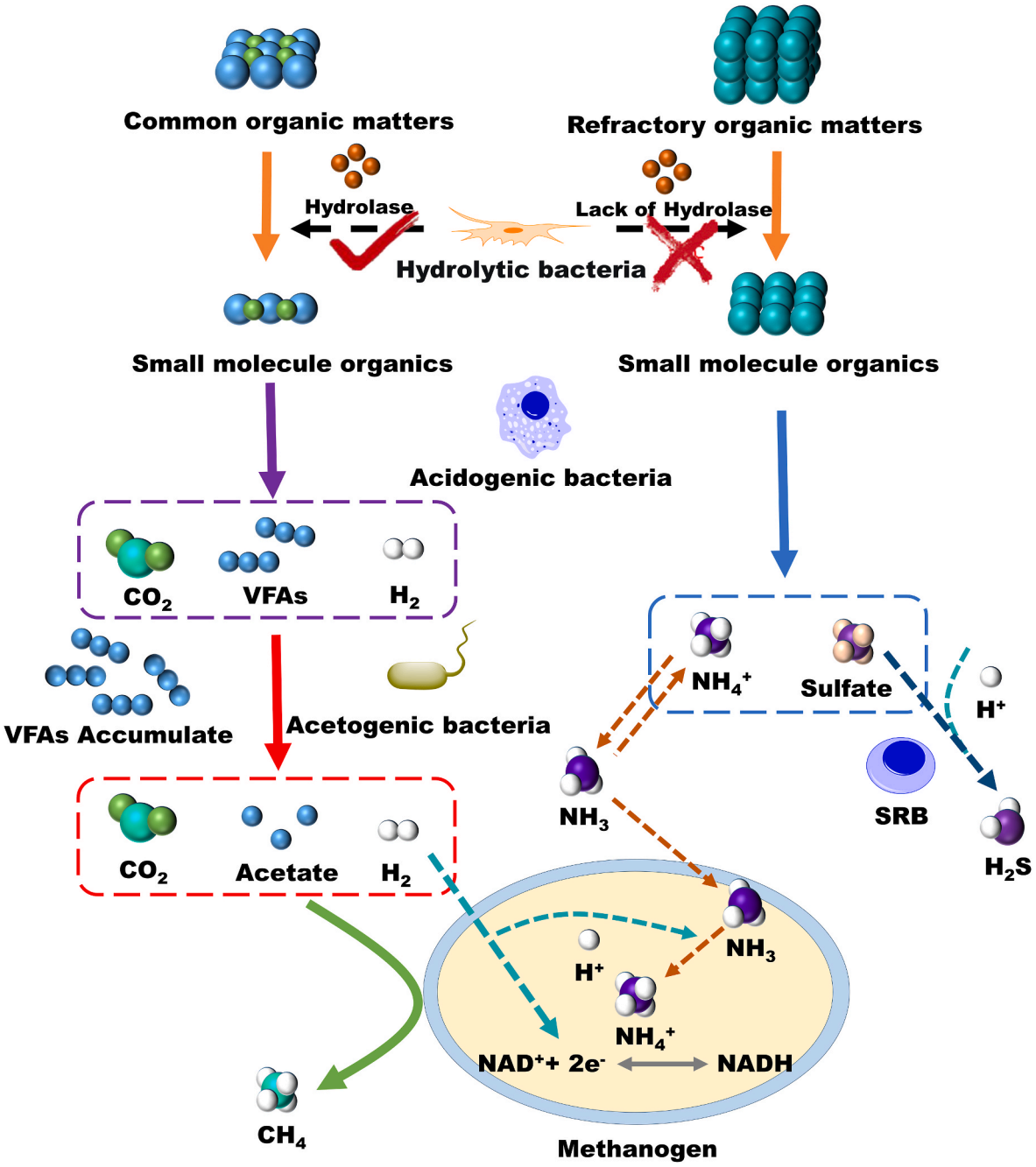


Fig. 3. Obstacles to AD for the treatment of industrial organic wastewater.

neutralization potential of AD process. As shown in Fig. 3 Uncontrolled accumulation of intermediates and byproducts including VFAs sulfide and sulfate has been a severe issue that limits the methane production (Yuan and Zhu, 2016). As key intermediates present in the AD process, VFAs serve as the substrates of methanogens, ensuring the availability of methanogenesis. However, due to the different growth rates of acid-producing fermenters and methanogenic bacteria, VFAs tend to accumulate during the AD process, resulting in a drop in pH and affecting microbial activity (Li et al., 2020). When the level of VFAs is below 750 mg/L, the methane content in biogas is close to 49.8%. However, when VFAs increase to 1400 mg/L, the activity of methanogenic bacteria is significantly suppressed due to the inhibitory effect of VFAs and the methane content in biogas decreases to about 30.7% (Zhao et al., 2020).

Sulfides and sulfates in AD systems are typically sourced from the degradation of sulfur-containing organic matters such as proteins or amino acids (Alex Kibangou et al., 2022). It should be noted that sulfate-reducing bacteria (SRB) in AD systems are highly resistant to low pH and will compete with methanogens for hydrogen sources to reduce sulfate (Zhao and Zhao, 2022). The sulfide (H_2S) produced during the sulfate reduction process is also toxic to methanogens and other microorganisms (Wu et al., 2018). Moreover, as the calorific value of combustion of biogas produced by AD is largely dependent on methane, the high content of H_2S significantly reduces the energy recovery potential of biogas (Mulu et al., 2021). As a toxic acid gas, H_2S will also corrode pipes and instruments (Zhou et al., 2016). Therefore, technical measures are necessary for mitigating the accumulation of intermediates and by-products, so that the AD system can be stabilized and energy recovery production can be increased.

4.3. Residual ammonia nitrogen

Ammonia nitrogen at relatively low concentrations (50–200 mg/L) can be used as a nitrogen nutrient for microbial growth and can also provide some alkalinity to maintain a suitable pH (Jiang et al., 2019). However, as shown in Fig. 3 high concentrations of ammonia nitrogen allows the presence of free ammonia nitrogen (FAN), which will greatly inhibit microbial activity in the AD process. Excessive FAN inhibits acetate metabolism and leads to the accumulation of other VFAs, which indirectly affects the process of AD (Liu et al., 2021). In addition, FAN is permeable to cell membranes and can enter methanogenic bacteria, and therefore can interfere with the metabolic pathway of methanogenic bacteria by inhibiting specific methane synthase reactions (Yan et al., 2019). In addition, due to the microbial action during AD, large amounts of organic nitrogen are converted into inorganic ammonia nitrogen, forming anaerobic digestion solution with high concentration of ammonia nitrogen (Mai et al., 2021; D. D. Yang et al., 2022). Therefore, in addition to alleviating the inhibitory effect of FAN, post-treatment of residual ammonia nitrogen is necessary.

5. Improvement and enhancement for AD of industrial organic wastewater

5.1. Micro-aeration pretreatment

Micro-aeration pretreatment involves the introduction of small amounts of oxygen or air into the AD system to form a special environment where aerobic and anaerobic are present alternately (Rashvanlou et al., 2021). The oxygen provided by micro-aeration is fully utilized by facultative anaerobic bacteria (such as hydrolysis bacteria, acidogenic bacteria, acetogenic bacteria) without inhibiting the activity of strict anaerobic bacteria such as methanogens (Zhou et al., 2021). In the presence of trace oxygen, the activity and community diversity of facultative anaerobic bacteria are significantly enhanced (Nguyen et al., 2019). The hydrolytic bacteria produce more extracellular hydrolases, which significantly improves the hydrolysis efficiency of refractory organic matter. This means that increase in the energy potential of industrial organic wastewater. At the same time, acidogenic bacteria and acetogenic bacteria promote the rapid production and degradation of VFAs, and consequently alleviate the inhibition of VFAs on AD system (Nguyen and Khanal, 2018). The presence of oxygen inhibits activity of SRB, increases sulfate oxidizing bacteria (SOB) activity, and sulfide is oxidized to sulfate, thiosulfate and elemental sulfur rather than being reduced to H_2S (Q. L. Chen et al., 2020; Krayzelova et al., 2015). Therefore, the micro-aeration pretreatment can remove H_2S from the AD system cost-effectively and efficiently. Table 4 demonstrates how micro-aeration pre-treatment is applied to improve the AD process. Besides, due to the extremely low energy consumption of micro-aeration and the increased methane production after micro aeration pretreatment in the AD system, the energy consumption of micro-aeration is completely offset by the energy recovered from AD. Kong et al. analyzed the energy consumption of the AD process using micro-aeration pretreatment and obtained a net energy consumption of −2.10 MJ/kg COD (Kong et al., 2019b). As show in Fig. 4, as an efficient and low energy consumption method for treating industrial organic wastewater, micro-aeration pretreatment shows the potential of carbon neutrality by promoting the hydrolysis of refractory organics, which alleviates VFAs inhibition and removing H_2S (Zhang et al., 2021).

5.2. Acidic and alkaline pretreatment

To solve the problem of slow hydrolysis efficiency of the AD process, pretreatment by adding acidic or alkaline chemical reagents is a promising method (Sołowski et al., 2020). The principle of acidic and alkaline pretreatment promoting AD process is shown in Fig. 4. Acidic and alkaline pretreatment can significantly destroy disrupt functional groups such as glycosidic bond, amide bond, and lactone bond in refractory organic matter, so that the hydrolysis efficiency can be boosted (Z. Chen et al., 2019). It also promotes the conversion of particulate organic matter to dissolved organic matter and thus increases methane yield (Li et al., 2015; Lu et al., 2018). Since the pH fluctuations of different types of industrial organic wastewater are relatively large, the addition of acidic or alkaline chemical reagents can balance the extreme pH of the wastewater. Acidic pretreatment makes hydrolase more effective by breaking down lignin and hemicellulose parts of biomass (Lorenci Woiciechowski et al., 2020). For acid pretreatment, sulfuric acid, nitric acid, hydrochloric acid, phosphoric acid are the most commonly used reagents (Loow et al., 2016). Alkaline pretreatments are more effective in removing lignin under milder conditions than acid pretreatments. Commonly used reagents for alkaline pretreatment are ammonia, sodium hydroxide, sodium carbonate, calcium hydroxide (Kim et al., 2016). The addition of alkaline chemical reagents will also increase the alkalinity of AD system and reduce the influence of VFAs accumulation on the system. Meanwhile, bicarbonate will promote the activity of methanogens and increase methane production (Almomani and Bhosale, 2020). The application of acid and alkaline pretreatment in improving AD process is shown in Table 4. In addition, the concept of carbon neutrality is particularly concerned with cost and energy consumption. Acidic and alkaline pretreatment is more energy efficient than physical or advanced oxidation processes. It is reported that the net energy of oxalic acid-ultrasonic pretreatment was positive (78.61 kJ/d), while the net energy of the control group and ultrasonic pretreatment was

Table 4
A list of micro-aeration and acid/alkaline pre-treatment methods used in anaerobic digestion of industrial organic wastewater.

Wastewater categories	Method	Condition	Improvements by the pre-treatment	Reference
Petrochemical wastewater	Micro-aeration	DO: (0.2–0.3 mg/L)	BOD_5/COD improved to 0.43; Avoid producing toxic H_2S.	Wu et al. (2015)
Pharmaceutical wastewater	Micro-aeration	DO: (0. - 1 mg/L)	BOD_5/COD improved by 0.18.	Chen et al. (2012)
Polymeric resins wastewater	Micro-aeration	DO: (0.6 and 1 mg/L)	COD removal rates improved to 80 ± 2%	Durán et al. (2008)
Industrial park wastewater	Micro-aeration	DO: (0.2–0.5 mg/L)	BOD_5/COD and COD removal rates increased to 0.61 and 43.8%.	An et al. (2020)
Synthetic protein wastewater	Acid: 4 M HCl	Room temperature & 6–24 h	When pH is 3, the methane yield increased by 29.4%.	(J. L. Li et al., 2022)
	Alkali: 4 M NaOH	Room temperature & 6–24 h	When pH is 12, methane yield increased by 35.7%.	
Pulp wastewater and spent coffee grounds	Acid: 1.0 M H_2SO_4	60 °C & 1 h	Methane production is 8 times of the control group.	Shen et al. (2022)
Starch mill wastewater	Alkali: 20 M NaOH	200 °C & 45 min	When pH is 12, SCOD improved to about 72000 mL/g.	Pyae et al. (2018)
	Acid: 36 N H_2SO_4	200 °C & 45 min	When pH is 2, sCOD improved to about 105000 mL/g.	
Food wastewater	Alkali: 2 N NaOH	Room temperature & 24 h	SCOD improved to 4500 ± 360 mL/g.	(Jallouli et al., 2021)
	Acid: 2 N H_2SO_4	Room temperature & 24 h	SCOD improved to 3999 ± 319.99 mL/g.	

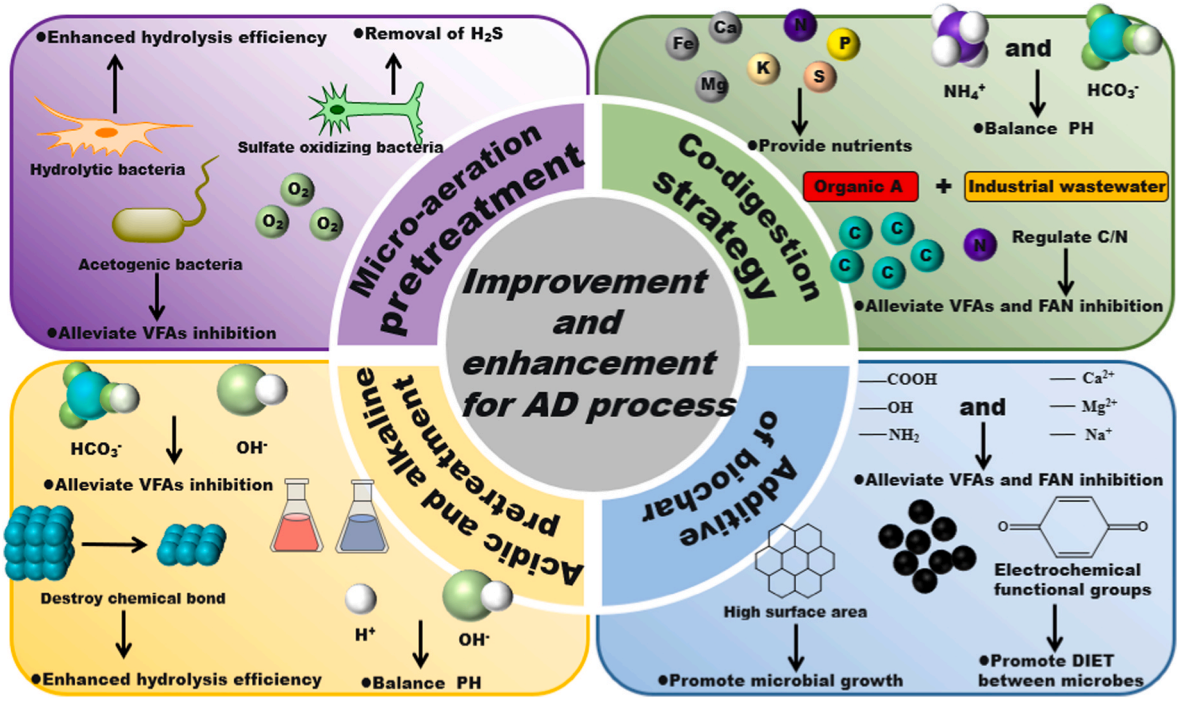


Fig. 4. Some methods to improvement and enhancement for AD process, including: Micro-aeration pretreatment; Acidic and alkaline pretreatment; Co-digestion strategy; Additive of biochar.

−121.91 kJ/d and −49.53 kJ/d, respectively. Through the cost analysis, only the net profit of oxalic acid-ultrasonic pretreatment reached 8.15 USD/ton, while the cost of control group and ultrasonic pretreatment was −131.8 USD/ton and −64.5 USD/ton (Gondi et al., 2023).

5.3. Co-digestion strategy

Co-digestion strategy involves the addition of external solid or liquid organic matters to industrial organic wastewater to stabilize the AD process and increase biogas production (Paranjpe et al., 2023). As a carbon neutral technology, co-digestion strategy has increased the energy self-sufficiency rate of sewage treatment plants to 80% (Budych--Gorzna et al., 2016). The AD process requires an appropriate C/N ratio. Industrial organic wastewaters from the starch and food processing industries have higher C/N and are more prone to VFAs accumulation, while wastewaters from slaughterhouse and ethanol processing industry have lower C/N content and are more susceptible to FAN inhibition (Siddique and Wahid, 2018; Xu et al., 2018). Therefore, As show in Fig. 4 the introduction of nitrogen-rich biomass (such as food waste and animal feces) or carbon-rich biomass into wastewater lacking carbon or nitrogen can effectively regulate the C/N of wastewater and alleviate VFAs and FAN inhibition (L. X. Li et al., 2019). Most organic wastes and bio-materials are highly nutritious and can provide nutrients necessary for microbial growth when used as substrates for co-digestion (Kunatsa and Xia, 2022). The addition of sewage sludge or animal manure can provide alkalinity and enhance the buffer capability of the AD system (Karki et al., 2021). In addition, the introduction of some easily degradable organic matter into the AD system makes easier to act as a substrate for hydrolytic bacteria, promoting the growth of hydrolytic bacteria and improving the degradation ability of refractory organic matter (Oduor et al., 2022). As indicated in Table 5, The co-digestion strategy has been applied to the treatment of industrial organic wastewater and has achieved good energy production. It is worth noting that the profits and energy output by the co-digestion strategy are far greater

Table 5
Co-digestion strategy used in the treatment of industrial organic wastewater and enhancement effect on methane production.

Wastewater Categories	Co-digestion substrate	Mix ratio	C/N	Temperature (°C)	OLR (kg VS/ m^3/d)	HRT (day)	Biogas yield (mL/g VS)	CH_4 content (%)	Reference
Olive mill wastewater	Cheese whey and liquid cow manure	55:40:5	*N.A*[a]	37	2.01	20	250.75	59.43	Dareioti and Kornaros (2014)
Olive mill wastewater	Swine manure	40:60	*N.A*	35 ± 1	1.2–4.4	30	278–373[b]	–	Kougias et al. (2014)
Dairy wastewater	Cattle manure	80:20	24.7	37	2.42 ± 0.1	20	870 ± 20	69.2 ± 1.5	Toumi et al. (2015)
Olive mill wastewater	Abattoir wastewater	40:60	28.21	55	4.1–12[c]	3.33–10	520 ± 70[d]	70–76	Gannoun et al. (2016)
Brewery wastewater	Piggery effluent	60:40	N.A	37 ± 1	2.22–4.96	3–6.7	196–238[b]	–	Eusébio et al. (2021)
Olive mill wastewater	Food waste	20:80	N.A	37	2.0 ± 0.1	N.A	302.16 ± 3.04[b]	–	El Gnaoui et al. (2020)

[a] Some studies did not provide the exact data.
[b] Several studies did provide methane yield. Unit: mL CH_4/g VS.
[c] Unit: kg COD/m^3/d.
[d] Unit: mL/g COD.

than its costs and energy consumption. It is report that co-digestion strategy increased power generation by 0.11 ± 0.03 kWh/m^3, get net profit 1.47€/PE/a and increased the energy self-sufficiency of WWTP by 16% (Macintosh et al., 2019).

5.4. Additives of biochar

Biochar is a kind of carbonaceous solid material obtained by pyrolysis of biomass under anaerobic or anoxic conditions above 250 °C (Kamali et al., 2021). As show in Fig. 4, the high specific surface area and porous structure of biochar can provide a suitable habitat for the growth and reproduction of microorganisms and form biofilms on the surface of biochar, and as a result, enhance the inhibitory effect of microorganisms on the accumulation of VFAs and FAN that may occur during the AD process (Codignole Luz et al., 2018). The addition of biochar enriches the community of hydrolytic bacteria and increases the release level of hydrolytic enzymes, and to a certain extent, also promotes the degradation efficiency of refractory organic matter (Duan et al., 2019). Besides, the existence functional groups of phenolic (X–OH), carboxylic (-COOH) groups, amine (-NH_2) groups and high content of Ca^{2+}, Mg^{2+}, Na^+, K^+ and other metal ions from biochar (Zhao et al., 2021), biochar also has a strong buffer capacity, which can effectively alleviate the inhibition effect of high concentration of VFAs and FAN on AD system. The presence of some electrochemical functional groups such as quinones and semiquinones can stimulate the direct interspecies electron transfer (DIET) of fermentation bacteria and methanogens, and significantly increase the degradation rate of VFAs and methane production (Wang et al., 2018, 2021). Under the concept of carbon emission reduction and energy recovery, the additional energy recovered by the AD processes through adding biochar resulted in a net gain of 0.7122 kWh/m^3 (Ni et al., 2023). According to the calculation, it can reduce carbon emissions by 0.37 kg CO_2/m^3 (Ang and Su, 2016). Some other examples of treatment of industrial organic wastewater by biochar enhanced AD process are shown in Table 6.

6. Post-treatment for nitrogen removal

Treating industrial organic wastewater with the AD process can lead to significant emissions of ammonia nitrogen, and this is another concern. The environmentally friendly and low energy anammox process has received widespread attention from researchers as a carbon neutral biological nitrogen removal technology from industrial organic wastewater (Zuo et al., 2023). Anammox bacteria (AnAOB) use nitrite to oxidize ammonium under anaerobic conditions, simplifying the complex process of traditional nitrogen removal (L. Zhang et al., 2022). Due to the significant reduction in aeration energy consumption and external carbon source requirements, the anammox process consumes only 0.9 kWh/kg N, which is much less than the energy consumption of the conventional nitrification/denitrification process which consumes 3.5 kWh/kg N (Trinh et al., 2021). Compared to conventional nitrification/denitrification process, anammox process reduces carbon emissions by 1.352 kg CO_2/kg N (Ang and Su, 2016). The carbon emissions decreased by 1.352 kg CO_2/kg N. In practice, AnAOB often interacts with ammonia oxidizing bacteria (AOB), nitrite oxidizing bacteria (NOB), denitrifying bacteria (DNF) for combined nitrogen removal (Huang et al., 2022), such as the partial nitrification/anammox process (PN/A) and the partial denitrification/anammox process (PD/A) (Ma et al., 2016), to achieve higher nitrogen removal rate. Biological nitrogen removal processes inevitably produce nitrous oxide, which has a more greenhouse effect than carbon dioxide. The nitrous oxide produced by conventional nitrification/denitrification process ranged from 0.008 to 0.067 g N_2O/g N, while the nitrous oxide produced by PN/A process and PD/A process ranged from 0.004 to 0.066 g N_2O/g N and 0.002–0.022 g N_2O/g N, respectively (Liu, 2023). Therefore, compared with traditional nitrification/denitrification, anammox is more friendly to the environment. The application of anammox and its combined processes for nitrogen removal from industrial organic wastewater are shown in Table 7. The PN/A process oxidizes part of the ammonium to nitrite through AOB, and the residual nitrate produced by the anammox process can also be returned to the AD process as an oxidant to promote COD degradation (Fan et al., 2022). The PD/A process completely oxidizes ammonia nitrogen to nitrate, followed by denitrification to produce nitrite. Although this process solves the difficulty of nitrite accumulation, it is slightly more energy intensive than the PN/A process (Sheng et al., 2020a). The specific microbial effects and associated stoichiometric relationships are shown in Fig. 5. However, some improvements to the anammox process are still needed to improve the shortcomings of AnAOB, such as high sensitivity to the environment and long doubling time (Q. Zhang et al., 2022). To improve the resistance of AnAOB to the external environment and the rapid start-up, the addition of biofilms or carriers is a viable approach (Jiang et al., 2020). However, when treating industrial organic wastewater under the concept of carbon neutrality, we prefer to couple a variety of carbon neutrality technologies to remove organic and inorganic pollutants in wastewater at the same time, and achieve a higher carbon neutrality potential under the premise of complementary technologies.

7. Future perspectives and outlooks

The energy-saving advantages of AD process in treating industrial organic wastewater and the upgraded technology and derivatives currently used to enhance the AD process should be the future concern. New technologies focusing on improving degradation performance and enhancing energy production and carbon reduction are highly

Table 6
Related studies on the promotion effect of biochar addition on anaerobic digestion of industrial organic wastewater and the enhancement of methane production.

Wastewater categories	Biochar feedstock	Preparation conditions		Dosage (g/L)	COD (mg/L)	Enhancement on removal performance	CH_4 yield improvement (%)	Reference
		Temperature (°C)	Time (h)					
Coal gasification wastewater	Sewage sludge[b]	700	3	5	2500–3000	COD removal rates increased by 25.4%.	68.1%	Zhuang et al. (2018)
Coal gasification wastewater	Water hyacinth[b]	700	2	5	2000	Removal rates of 2,4,6-trichlorophenol and COD improved to 98.8% and 98.9%.	From 1.25 L to 2 L	Zhuang et al. (2020)
Ammonium wastewater	Fruitwood	800–900	N.A[a]	10	N.A	N.A	47.1%	Lü et al. (2016)
Phenolic wastewater	Wood	450	0.5	5	N.A	Phenol removal rates improved to 86.2%.	17.6%	Pan et al. (2021)
Phenolic wastewater	loofah sponge[b]	600	2	5	1050–2550	2,4,6-trichlorophenol removal rates × 1.81; COD removal rates × 2.1.	× 1.65	Xie et al. (2022)
Pharmaceutical wastewater	Coconut[b]	600	4	N.A	1600–2000	Removal rates of sulfamethoxazole and COD increased by 36.51% and 22.76%.	× 6.19	Ni et al. (2023)

[a] Some studies did not provid the exact data.
[b] Some studies did used modified biochar.

Table 7
Application of Anammox process and the integrated combination process for biological nitrogen removal from industrial organic wastewater.

Wastewater categories	Anamomox type	Integrated process	Temperature (°C)	HRT (days)	NLR (g N/m^3/d)	Influent concentrations (mg/L)			TNRE (%)	TNRR (g/(m^3·d))	Reference
						NH_4^+-N	NO_2^--N	NO_3^--N			
Semiconductors wastewater	PD/A	Anaerobic reactor	33 ± 2	0.25	*N.A*[a]	410	*N.A*	610	98	4110	(X. Y. Li et al., 2019)
Optoelectronics wastewater	PN/A	SBR	25	4	909	3636	*N.A*	*N.A*	89 ± 3	791	Daverey et al. (2013)
Semiconductor wastewater	PN/A	SBR	Room temperature	4	665	2660	*N.A*	*N.A*	75.7	506	Liang et al. (2016)
Pharmaceuticals wastewater	PD/A	SBR	Room temperature	20[b]	*N.A*	77.9 ± 2.6	*N.A*	104.1 ± 4.4	81.2 ± 6.3	*N.A*	(J. Zhang et al., 2022)
Slaughterhouse wastewater	PN/A	UASB	37 ± 1.1	1.0 ± 0.2[b]	6300 ± 800	82.9 ± 20.3	137.5 ± 24.3	43.7 ± 7.9	87.7 ± 5.6	5500 ± 800	Silveira et al. (2021)
Coal to ethylene glycol wastewater	PN/A	IFAS	31 ± 2	19–36[b]	410–780	300–800	2.4–240	*N.A*	77.3 ± 3.3	430–50	Yao et al. (2022)
Fish canning wastewater	PN/A	SBR	24 ± 2	1.3 ± 0.1	NA	220 ± 35	*N.A*	*N.A*	87	180 ± 20	Val Del Rio et al. (2018)

[a] Some studies did not provid the exact data.
[b] Unit: hours.

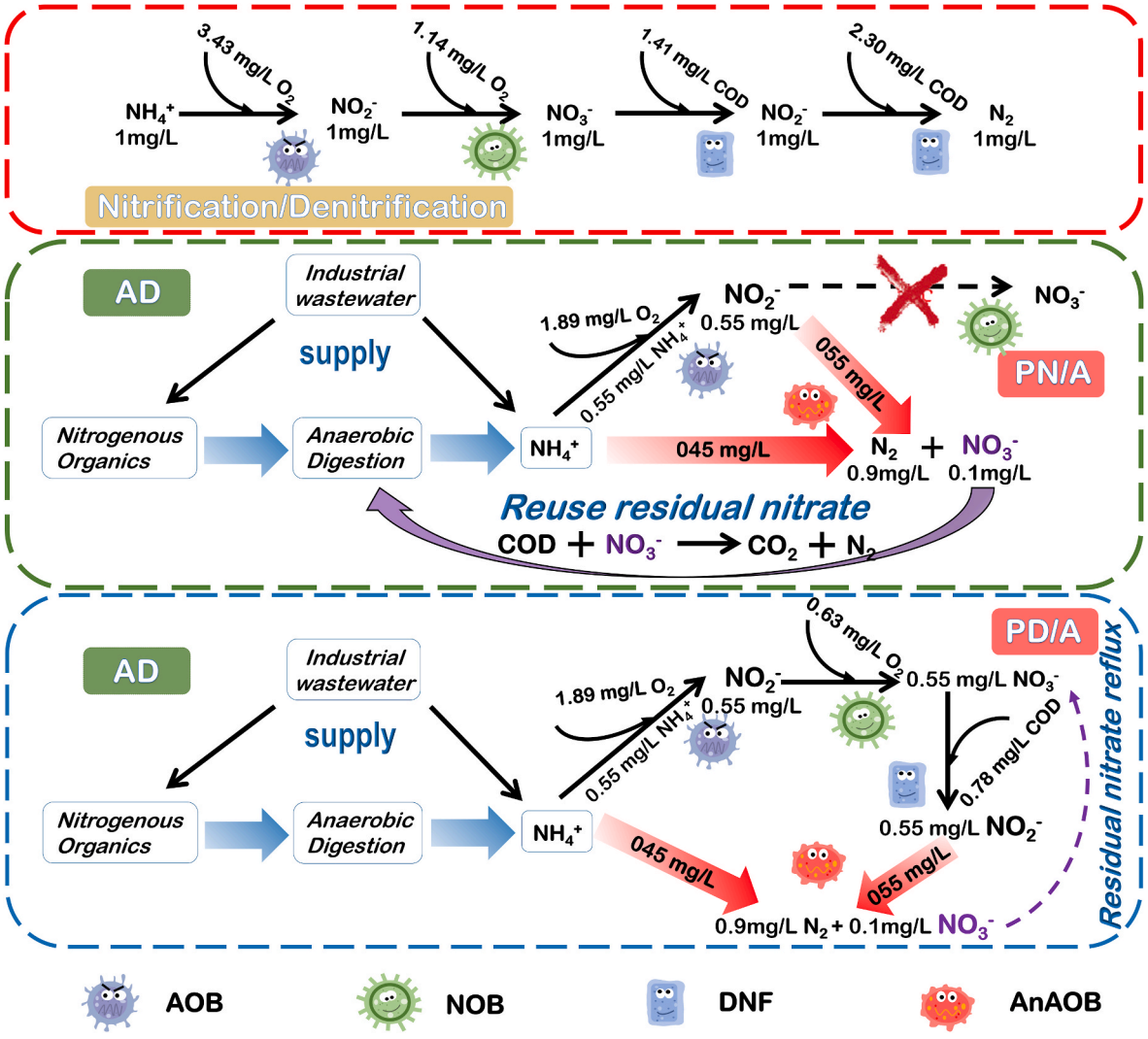


Fig. 5. The interaction and stoichiometric relationship between microorganisms in nitrification/denitrification, PN/A processes and PD/A processes.

recommended. Micro-aeration pretreatment, acidic and alkaline pretreatment, co-digestion strategy and addition of biochar have significantly improved the stability and energy yield of AD systems at low energy consumption or cost. However, AD technology is not the only way to achieve carbon-neutral treatment of industrial organic wastewater. The other bio-technologies with the potential to be carbon-

neutral should be further investigated in the future. Therefore, we propose bio-technologies combined with AD and discuss in detail their feasibility and carbon neutrality in the treatment of industrial organic wastewater.

7.1. AnMBR and anammox integrated process

In recent years, the process integrated AnMBR with partial-nitritation anammox (PN/A) is attracting attention.For AnMBR, anammox removes high concentration of ammonium from the AnMBR leachate (Guo et al., 2022). Besides, AnMBR can effectively remove COD from water and transform organic nitrogen into ammonium. Its membrane components can effectively intercept heterotrophic microorganism and suspended solids, which optimizes the external environment of the AnAOB (Wu et al., 2021). Because of the excellent quality of AnMBR effluent, the residual SCOD in effluenti is suitable as carbon source to promote the possible denitrification process in PN/A. It has been reported that the PN/A process is able to remove 50%–70% of COD

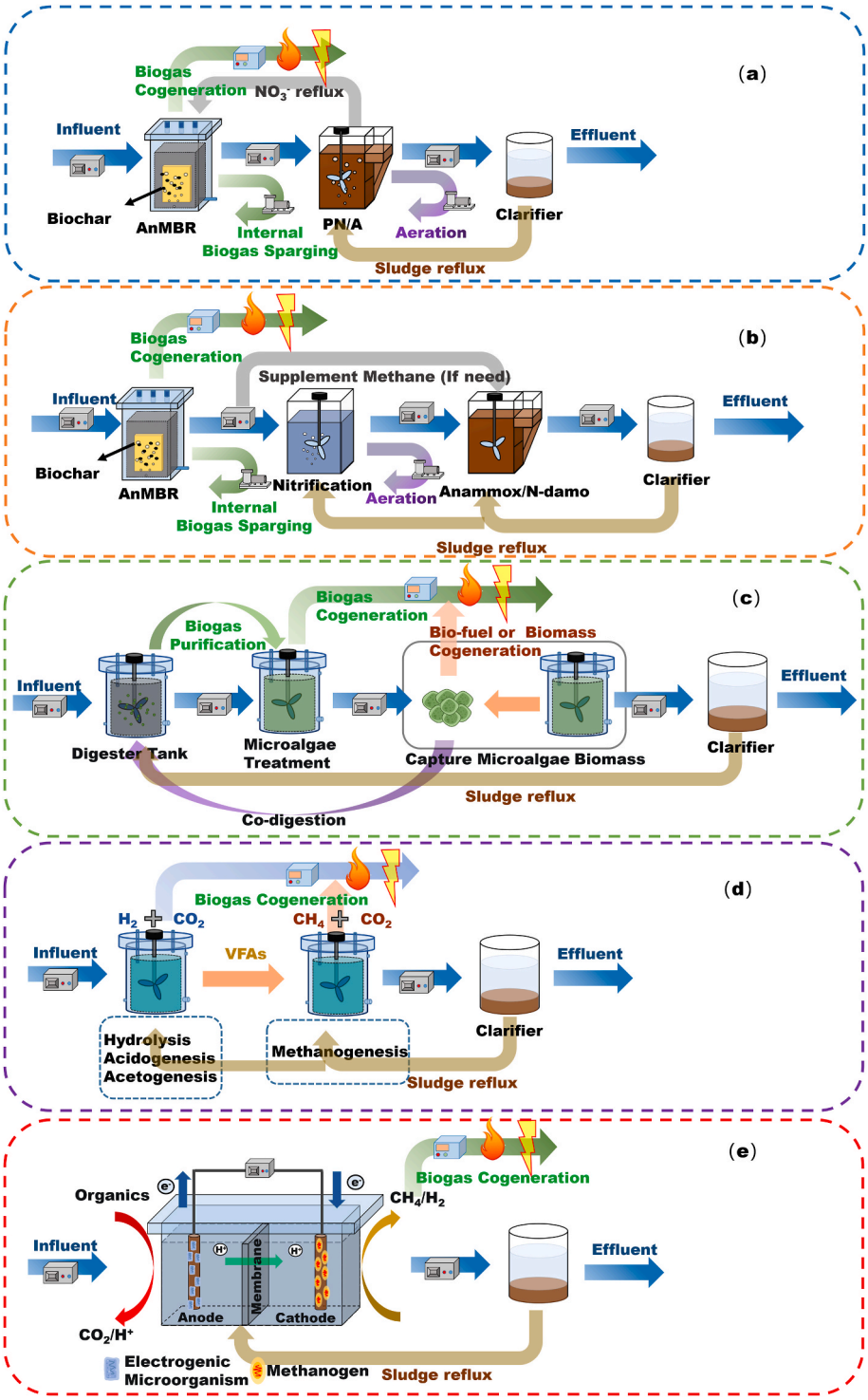


Fig. 6. Future perspectives and outlooks, including: (a) AnMBR/PN/A process; (b) AnMBR/Anammox/n-damo process; (c) AD combined with microalgae culture; (d) Two-stage AD to produce hydrogen and methane; (e) Bioelectrochemical system based on AD.

from AD effluent (Sheng et al., 2020b). Therefore, coupling AnMBR and PN/A processes can achieve more efficient COD and TN removal efficiency through synergistic action. The AnMBR/PN/A process has been reported to achieve 95.1% and 81.7% removal of COD and TN respectively when treating municipal wastewater(Rong et al., 2021). Compared with the traditional A^2/O process, the combination of AnMBR/PN/A process can reduce the energy consumption by 0.416 kWh/m^3 and CO_2 emission by 74.9% (Kong et al., 2022d). It is conceivable that the integrated AnMBR and anammox processes have a higher reduction in carbon emissions and energy consumption potential for the treatment of industrial organic wastewater. In practice, methane produced by AnMBR is largely dissolved in wastewater and cannot be directly used as energy (Sanchis-Perucho et al., 2021), and methane is a more powerful greenhouse gas than carbon dioxide (Black et al., 2021). Rather than consuming energy to extract dissolved methane from wastewater, it would be better to use the methane in this form directly. Therefore, nitrate-driven anaerobic methane oxidation (N-damo) is becoming the focus of future research (Zhen et al., 2019). The process solves the problem of dissolved methane and additional nitrate production, while also providing nitrite for anammox (Haroon et al., 2013). The N-damo process eliminates the adverse effects of NOB. Since the nitrification process is complete (from ammonium to nitrate), it is easier to control the aeration process, however, it requires, at the same time, more energy to supply oxygen. Compared with A^2/O process, the AnMBR/Anammox/n-damo system reduces costs by at least $ 0.300/d/m^3 and greenhouse gas emissions by 1.65 kg CO_2/d/m^3, respectively (Cogert et al., 2019). The process of wastewater treatment by AnMBR and anammox integrated process is shown in Fig. 6 (a) and **(b).**

However, the treatment of membrane fouling consumes 60%–75% of the total energy consumption of AnMBR reactor, and the problem of membrane fouling is still a major obstacle for the coupling process to achieve carbon neutrality (Pretel et al., 2014). The studies have shown that when AnMBR is used to treat industrial wastewater with COD of 3219 mg/L, the energy consumption of membrane fouling (0.46 kWh/m^3) accounts for 93% of the total energy consumption (0.49 kWh/m^3) and 13% of the energy production by AnMBR (3.45 kWh/m^3) (Kong et al., 2022c). To deal with membrane fouling with low energy consumption, the use of biochar modified AnMBR is a preferred option (Nabi et al., 2023). Biochar can be used without activation and its specific gravity is lower than the total specific gravity, which means that the energy consumption required for fluidization is also low (Cheng et al., 2021; Lei et al., 2021). The presence of AnMBR membrane components makes it easier to recover the biochar in granular form. The absorption of SMP and EPS and the removal of toxic pollutants by biochar are the major reasons for the reduction of membrane pollution (Cheng et al., 2021; Ghaffar et al., 2018). From the perspective of microorganisms, the massive aggregation of genus *Mesotoga* on the biochar surface promotes the decomposition of SMPs and EPS, especially the breakdown of the macromolecular protein EPS, which relieves the pressure of membrane contamination (L. Q. Chen et al., 2020). In conclusion, the addition of biochar to AnMBR can simultaneously enhance the AD process and alleviate membrane contamination of AnMBR. When biochar is added to AnDMBR, total membrane filtration resistance is reduced by 6.2 × 10^{11}–7.3 × 10^{11} m^{-1}, and methane production is increased by 12% (Pan et al., 2023).

7.2. AD combined with microalgae culture

Microalgae can absorb N, P or organic carbon under autotrophic or heterotrophic conditions, and, in addition, they are characterized by their special ability to fix carbon dioxide and release oxygen (Ahmad et al., 2022). The combination of microalgae culture and AD process is likely to be the future carbon neutral technology for the treatment of industrial organic wastewater. As shown in Fig. 6 (c), in this technology, microalgae utilize the substances (COD, TN, TP and VFAs) in the AD filtrate to supply their own growth while purifying biogas by capture CO_2 (W. L. Yang et al., 2022). It is report that Treatment of AD filtrate with microalgae has been reported to remove 39.02–68.11% of COD, 26.58–59.08% of TN, 22.78–60.03% of TP and the methane content in biogas has increased to 73.61–84.28% (Wang et al., 2016). In addition, mature microalgae are rich in carbohydrates, proteins, fats and a variety of nutrients and are an excellent co-digestive substrate (Kim and Kang, 2015). Excess microalgal biomass can be adopted to produce high-quality bio-fuels, which is more environmentally friendly than traditional fossil fuels (Hoh et al., 2016). Biomass and bio-fuels can be burned to generate electricity, further improving energy recovery efficiency. It is report that bio-fuels and biomass burned to generate electricity can offset 22–50% of the energy consumption of WWTP (Sturm and Lamer, 2011). Herefore, AD combined with microalgae culture can enhance the carbon neutral potential of the AD process by purifying biogas, providing co-digestive substrate, and burning biomass or bio-fuels for power generation.

7.3. Two-stage AD to produce hydrogen and methane

It is well known that the products of both the acidogenesis and acetogenesis of the AD process contain hydrogen and hydrogen is a renewable, clean energy source (whose combustion produces only water vapor) with a high energy density (142 MJ/kg) (Mishra et al., 2018). Therefore, as shown in Fig. 6 (d), the AD process is divided into hydrogen and methane generation systems to recover both hydrogen and methane and to increase the carbon neutral potential of industrial organic wastewater treatment. In the first stage, complex organic matter is transformed into VFAs in the presence of different microorganisms, while hydrogen is generated. In the second stage, methanogens convert VFAs produced at the first stage into methane (Hou et al., 2021). Two-stage AD can provide suitable growth environment for microorganisms in hydrogen and methanogenic systems respectively, prevent VFAs accumulation caused by increasing hydrogen partial pressure, and improve energy recovery efficiency (Kim et al., 2022). The two-stage AD process is reported to have a 12.9% higher energy recovery for hydrogen and methane production than the single-stage AD process (Fu et al., 2017). High yields of hydrogen (101 mL H_2/g VS) and methane (629 mL H_2/g VS) have been reported for the treatment of rubber sheet wastewater with the two-stage AD process (Promnuan et al., 2020).

7.4. Bioelectrochemical system based on AD

Bioelectrochemical system (BES) is a flexible and sustainable technology that can be applied in combination with technologies such as AD. Common BES include microbial fuel cell (MFC), microbial electrolytic cell (MEC), microbial desalting cell (MDC), and microbial electrosynthesis (MES) (W. W. Wang et al., 2022). The integration of BES with AD (BES-AD) can operate with high energy efficiency, increasing the potential for carbon neutrality on the basis of AD (Beegle and Borole, 2018). A common BES-AD system is shown in Fig. 6 (e). BES-AD system can promote the aggregation of electrochemically active bacteria, enhance the DIET between bacteria and methanogens, and thus promote the degradation of refractory substances and accelerate transformation of methane (Guo et al., 2017; Xie et al., 2021). The BES-UASB coupled system treated the Fischer-Tropsch wastewater, and the COD removal rate and methane production were 86.8% and 2.31 ± 0.1 L/L/d, respectively, as compared to 72.1% and 1.77 ± 0.08 L/L/d when using only the UASB system (Wang et al., 2017). The BES-AD system can effectively reduce the influence of H_2S on the AD system during the treatment of high strength sulfate industrial organic wastewater due to the creation of weak alkaline conditions at the cathode, and the methane production is 3.03 times that of the AD reactor without alkaline pH control (Yuan et al., 2020).

8. Conclusions

Industrial organic wastewater contains a lot of energy, and recycling energy from industrial waste water is an important way to achieve carbon neutral treatment of industrial waste water. This paper systematically reviews the methods, feasibility and potential of energy recovery of industrial organic wastewater. Here are the key conclusions.

(1) Anaerobic digestion (AD) offers significant energy advantages over traditional biological processes for treating industrial organic wastewater. However, in view of the shortcomings of anaerobic digestion, it is still necessary to improve and optimize anaerobic digestion. It is recommended to use micro-aeration pretreatment, acidic and alkaline pretreatment, co-digestion and adding biochar to improve anaerobic digestion.
(2) For industrial organic wastewater with high ammonia nitrogen, nitrogen removal treatment is also needed after anaerobic digestion. It is recommended to use AnMBR and Anammox process coupling to conduct nitrogen removal treatment of industrial organic wastewater, because this process is more energy saving and has higher energy recovery efficiency.
(3) For AD process combined with biological hydrogen production, microalgae culture, bioelectrochemical technology and AD combined technology broaden the ideas for the future industrial organic wastewater carbon neutral treatment.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

The authors do not have permission to share data.

Acknowledgments

This work was supported by the Natural Science Foundation of Jiangsu Province (No. BK20210871), National Natural Science Foundation of China (No. 52100100), Suzhou Innovation and Entrepreneurship Leader Project (No. ZXL2022498), Graduate Practice and Innovation Project of Jiangsu Province (No. SJCX23_1719) and Jiangsu Foreign Expert Workshop of Suzhou University of Science and Technology.

References

Ahmad, A., Banat, F., Alsafar, H., Hasan, S.W., 2022. Algae biotechnology for industrial wastewater treatment, bioenergy production, and high-value bioproducts. Sci. Total Environ. 806, 150585 https://doi.org/10.1016/j.scitotenv.2021.150585.

Alex Kibangou, V., Lilly, M., Busani Mpofu, A., de Jonge, N., Oyekola, O.O., Jean Welz, P., 2022. Sulfate-reducing and methanogenic microbial community responses during anaerobic digestion of tannery effluent. Bioresour. Technol. 347, 126308 https://doi.org/10.1016/j.biortech.2021.126308.

Almomani, F., Bhosale, R.R., 2020. Enhancing the production of biogas through anaerobic co-digestion of agricultural waste and chemical pre-treatments. Chemosphere 255, 126805. https://doi.org/10.1016/j.chemosphere.2020.126805.

An, Y., Wang, M., Zhou, Z., Sun, X., Cheng, C., Zhu, S., Wang, L., Wu, Z., 2020. Enhancing biodegradability of industrial park wastewater by packing carriers and limited aeration in the hydrolysis process. J. Clean. Prod. 264, 121638 https://doi.org/10.1016/j.jclepro.2020.121638.

Ang, B.W., Su, B., 2016. Carbon emission intensity in electricity production: a global analysis. Energy Pol. 94, 56–63. https://doi.org/10.1016/j.enpol.2016.03.038.

Anjum, F., Khan, I.M., Kim, J., Aslam, M., Blandin, G., Heran, M., Lesage, G., 2021. Trends and progress in AnMBR for domestic wastewater treatment and their impacts on process efficiency and membrane fouling. Environ. Technol. Innov. 21, 101204 https://doi.org/10.1016/j.eti.2020.101204.

Aslam, A., Khan, S.J., Shahzad, H.M.A., 2022. Anaerobic membrane bioreactors (AnMBRs) for municipal wastewater treatment- potential benefits, constraints, and future perspectives: an updated review. Sci. Total Environ. 802, 149612 https://doi.org/10.1016/j.scitotenv.2021.149612.

Balcıoğlu, G., Yilmaz, G., Gönder, Z.B., 2021. Evaluation of anaerobic membrane bioreactor (AnMBR) treating confectionery wastewater at long-term operation under different organic loading rates: performance and membrane fouling. Chem. Eng. J. 404, 126261 https://doi.org/10.1016/j.cej.2020.126261.

Basset, N., Santos, E., Dosta, J., Mata-Álvarez, J., 2016. Start-up and operation of an AnMBR for winery wastewater treatment. Ecol. Eng. 86, 279–289. https://doi.org/10.1016/j.ecoleng.2015.11.003.

Beegle, J.R., Borole, A.P., 2018. Energy production from waste: evaluation of anaerobic digestion and bioelectrochemical systems based on energy efficiency and economic factors. Renew. Sustain. Energy Rev. 96, 343–351. https://doi.org/10.1016/j.rser.2018.07.057.

Black, J.L., Davison, T.M., Box, I., 2021. Methane emissions from Ruminants in Australia: mitigation potential and applicability of mitigation strategies. Animals 11, 951. https://doi.org/10.3390/ani11040951.

Budych-Gorzna, M., Smoczynski, M., Oleskowicz-Popiel, P., 2016. Enhancement of biogas production at the municipal wastewater treatment plant by co-digestion with poultry industry waste. Appl. Energy 161, 387–394. https://doi.org/10.1016/j.apenergy.2015.10.007.

Chen, L., Cheng, P., Ye, L., Chen, H., Xu, X., Zhu, L., 2020. Biological performance and fouling mitigation in the biochar-amended anaerobic membrane bioreactor (AnMBR) treating pharmaceutical wastewater. Bioresour. Technol. 302, 122805 https://doi.org/10.1016/j.biortech.2020.122805.

Chen, Q., Wu, W., Qi, D., Ding, Y., Zhao, Z., 2020. Review on microaeration-based anaerobic digestion: state of the art, challenges, and prospectives. Sci. Total Environ. 710, 136388 https://doi.org/10.1016/j.scitotenv.2019.136388.

Chen, R., Wen, W., Jiang, H., Lei, Z., Li, M., Li, Y.-Y., 2019. Energy recovery potential of thermophilic high-solids co-digestion of coffee processing wastewater and waste activated sludge by anaerobic membrane bioreactor. Bioresour. Technol. 274, 127–133. https://doi.org/10.1016/j.biortech.2018.11.080.

Chen, Z., Dou, X., Zhang, Y., Yang, M., Wei, D., 2019. Rapid thermal-acid hydrolysis of spiramycin by silicotungstic acid under microwave irradiation. Environ. Pollut. 249, 36–44.

Chen, Z.L., Liu, Z.Z., Sun, X.R., Huang, X.H., Zhan, J., 2012. A novel application of micro-aerobic hydrolysis and acidification on the treatment of refractory Chinese traditional medicine wastewater. Appl. Mech. Mater. 178–181, 570–574. https://doi.org/10.4028/www.scientific.net/AMM.178-181.570.

Cheng, D., Ngo, H.H., Guo, W., Chang, S.W., Nguyen, D.D., Nguyen, Q.A., Zhang, J., Liang, S., 2021. Improving sulfonamide antibiotics removal from swine wastewater by supplying a new pomelo peel derived biochar in an anaerobic membrane bioreactor. Bioresour. Technol. 319, 124160 https://doi.org/10.1016/j.biortech.2020.124160.

Cheng, S., Ran, X., Ren, G., Wei, Z., Wang, Z., Rao, T., Li, R., Ma, X., 2022. Comparison of Fenton and ozone oxidation for pretreatment of petrochemical wastewater: COD removal and biodegradability improvement mechanism. Separations 9, 179. https://doi.org/10.3390/separations9070179.

Codignole Luz, F., Cordiner, S., Manni, A., Mulone, V., Rocco, V., 2018. Biochar characteristics and early applications in anaerobic digestion-a review. J. Environ. Chem. Eng. 6, 2892–2909. https://doi.org/10.1016/j.jece.2018.04.015.

Cogert, K.I., Ziels, R.M., Winkler, M.K.H., 2019. Reducing cost and environmental impact of wastewater treatment with denitrifying methanotrophs, anammox, and mainstream anaerobic treatment. Environ. Sci. Technol. 53, 12935–12944. https://doi.org/10.1021/acs.est.9b04764.

Cruz-Salomón, A., Ríos-Valdovinos, E., Pola-Albores, F., Lagunas-Rivera, S., Cruz-Rodríguez, R.I., Cruz-Salomón, K., del, C., Hernández-Méndez, J.M.E., Domínguez-Espinosa, M.E., 2020a. Treatment of cheese whey wastewater using an expanded granular sludge bed (EGSB) bioreactor with biomethane production. Processes 8, 931. https://doi.org/10.3390/pr8080931.

Cruz-Salomón, A., Ríos-Valdovinos, E., Pola-Albores, F., Lagunas-Rivera, S., Cruz-Rodríguez, R.I., Cruz-Salomón, K.D.C., Hernández-Méndez, J.M.E., Domínguez-Espinosa, M.E., 2020b. Treatment of cheese whey wastewater using an expanded granular sludge bed (EGSB) bioreactor with biomethane production. Processes 8, 931. https://doi.org/10.3390/pr8080931.

Dareioti, M.A., Kornaros, M., 2014. Effect of hydraulic retention time (HRT) on the anaerobic co-digestion of agro-industrial wastes in a two-stage CSTR system. Bioresour. Technol. 167, 407–415. https://doi.org/10.1016/j.biortech.2014.06.045.

Daverey, A., Su, S.-H., Huang, Y.-T., Chen, S.-S., Sung, S., Lin, J.-G., 2013. Partial nitrification and anammox process: a method for high strength optoelectronic industrial wastewater treatment. Water Res. 47, 2929–2937. https://doi.org/10.1016/j.watres.2013.01.028.

Dereli, R.K., 2019. Modeling long-term performance of full-scale anaerobic expanded granular sludge bed reactor treating confectionery industry wastewater. Environ. Sci. Pollut. Res. 26, 25037–25045. https://doi.org/10.1007/s11356-019-05739-1.

Deschamps, L., Merlet, D., Lemaire, J., Imatoukene, N., Filali, R., Clément, T., Lopez, M., Theoleyre, M.-A., 2021. Excellent performance of anaerobic membrane bioreactor in treatment of distillery wastewater at pilot scale. J. Water Process Eng. 41, 102061 https://doi.org/10.1016/j.jwpe.2021.102061.

di Cicco, M.R., Spagnuolo, A., Masiello, A., Vetromile, C., Nappa, M., Corbo, G., Lubritto, C., 2019. Assessing energy performance and critical issues of a large wastewater treatment plant through full-scale data benchmarking. Water Sci. Technol. 80, 1421–1429. https://doi.org/10.2166/wst.2019.392.

Drewnowski, J., Remiszewska-Skwarek, A., Duda, S., Łagód, G., 2019. Aeration process in bioreactors as the main energy consumer in a wastewater treatment plant. Review of solutions and methods of process optimization. Processes 7, 311. https://doi.org/10.3390/pr7050311.

Duan, X., Chen, Yunzhi, Yan, Y., Feng, L., Chen, Yinguang, Zhou, Q., 2019. New method for algae comprehensive utilization: Algae-derived biochar enhances algae anaerobic fermentation for short-chain fatty acids production. Bioresour. Technol. 289, 121637 https://doi.org/10.1016/j.biortech.2019.121637.

Durán, U., Monroy, O., Gómez, J., Ramírez, F., 2008. Biological wastewater treatment for removal of polymeric resins in UASB reactor: influence of oxygen. Water Sci. Technol. 57, 1047–1052. https://doi.org/10.2166/wst.2008.058.

El Gnaoui, Y., Sounni, F., Bakraoui, M., Karouach, F., Benlemlih, M., Barz, M., El Bari, H., 2020. Anaerobic co-digestion assessment of olive mill wastewater and food waste: effect of mixture ratio on methane production and process stability. J. Environ. Chem. Eng. 8, 103874 https://doi.org/10.1016/j.jece.2020.103874.

Eusébio, A., Neves, A., Marques, I.P., 2021. Complementary substrates-Brewery wastewater and Piggery effluent—assessment and microbial community Profiling in a hybrid anaerobic reactor. Appl. Sci. 11, 4364. https://doi.org/10.3390/app11104364.

Fan, Y., Tan, X., Huang, Y., Hao, T., Chen, H., Yi, X., Li, D., Pan, Y., Li, Y., Kong, Z., 2022. Chemical oxygen demand and nitrogen removal from real membrane-manufacturing wastewater by a pilot-scale internal circulation reactor integrated with partial nitritation-anammox. Bioresour. Technol. 364, 128116 https://doi.org/10.1016/j.biortech.2022.128116.

Faria, C.V., Souza, D.F., Pontes, T.M., Amaral, M.C.S., Fonseca, F.V., 2019. Strategies of anaerobic sludge granulation in an EGSB reactor. J. Environ. Manage. 244, 69–76. https://doi.org/10.1016/j.jenvman.2019.05.015.

Fu, S.-F., Xu, X.-H., Dai, M., Yuan, X.-Z., Guo, R.-B., 2017. Hydrogen and methane production from vinasse using two-stage anaerobic digestion. Process Saf. Environ. Protect. 107, 81–86. https://doi.org/10.1016/j.psep.2017.01.024.

Gannoun, H., Omri, I., Chouari, R., Khelifi, E., keskes, S., Godon, J.-J., Hamdi, M., Sghir, A., Bouallagui, H., 2016. Microbial community structure associated with the high loading anaerobic codigestion of olive mill and abattoir wastewaters. Bioresour. Technol. 201, 337–346. https://doi.org/10.1016/j.biortech.2015.11.050.

Garfí, M., Flores, L., Ferrer, I., 2017. Life Cycle Assessment of wastewater treatment systems for small communities: activated sludge, constructed wetlands and high rate algal ponds. J. Clean. Prod. 161, 211–219. https://doi.org/10.1016/j.jclepro.2017.05.116.

Ghaffar, A., Zhu, X., Chen, B., 2018. Biochar composite membrane for high performance pollutant management: fabrication, structural characteristics and synergistic mechanisms. Environ. Pollut. 233, 1013–1023. https://doi.org/10.1016/j.envpol.2017.09.099.

Gondi, R., Akila Ravi, Y.K., Kavitha, S., Al-Qaradawi, S.Y., Challabathula, D., Kumar, G., Rajesh Banu, J., 2023. Cost-effective bio-methanation via oxalic acid coupled sonication pretreatment of Gracilaria salicornia. Biomass Bioenergy 175, 106876. https://doi.org/10.1016/j.biombioe.2023.106876.

Gu, Y., Li, Y., Li, X., Luo, P., Wang, H., Robinson, Z.P., Wang, X., Wu, J., Li, F., 2017. The feasibility and challenges of energy self-sufficient wastewater treatment plants. Appl. Energy 204, 1463–1475. https://doi.org/10.1016/j.apenergy.2017.02.069.

Guo, J.-H., Peng, Y.-Z., Peng, C.-Y., Wang, S.-Y., Chen, Y., Huang, H.-J., Sun, Z.-R., 2010. Energy saving achieved by limited filamentous bulking sludge under low dissolved oxygen. Bioresour. Technol. 101, 1120–1126. https://doi.org/10.1016/j.biortech.2009.09.051.

Guo, T., Ji, Y., Zhao, J., Horn, H., Li, J., 2020. Coupling of Fe-C and aerobic granular sludge to treat refractory wastewater from a membrane manufacturer in a pilot-scale system. Water Res. 186, 116331 https://doi.org/10.1016/j.watres.2020.116331.

Guo, Y., Sanjaya, E.H., Rong, C., Wang, T., Luo, Z., Chen, H., Wang, H., Hanaoka, T., Sakemi, S., Ito, M., Kobayashi, S., Kobayashi, M., Li, Y.-Y., 2022. Treating the filtrate of mainstream anaerobic membrane bioreactor with the pilot-scale sludge-type one-stage partial nitritation/anammox process operated from 25 to 15 °C. Bioresour. Technol. 351, 127062 https://doi.org/10.1016/j.biortech.2022.127062.

Guo, Z., Liu, W., Yang, C., Gao, L., Thangavel, S., Wang, L., He, Z., Cai, W., Wang, A., 2017. Computational and experimental analysis of organic degradation positively regulated by bioelectrochemistry in an anaerobic bioreactor system. Water Res. 125, 170–179. https://doi.org/10.1016/j.watres.2017.08.039.

Han, J.-L., Xia, X., Haider, M.R., Jiang, W.-L., Tao, Y., Liu, M.-J., Wang, H., Ding, Y.-C., Hou, Y.-N., Cheng, H.-Y., Wang, A.-J., 2018. Functional graphene oxide membrane preparation for organics/inorganic salts mixture separation aiming at advanced treatment of refractory wastewater. Sci. Total Environ. 628–629, 261–270. https://doi.org/10.1016/j.scitotenv.2018.02.043.

Hao, X., Liu, R., Huang, X., 2015. Evaluation of the potential for operating carbon neutral WWTPs in China. Water Res. 87, 424–431. https://doi.org/10.1016/j.watres.2015.05.050.

Haroon, M.F., Hu, S., Shi, Y., Imelfort, M., Keller, J., Hugenholtz, P., Yuan, Z., Tyson, G. W., 2013. Anaerobic oxidation of methane coupled to nitrate reduction in a novel archaeal lineage. Nature 500, 567–570. https://doi.org/10.1038/nature12375.

Hassan, S.R., Hung, Y.-T., Dahlan, I., Abdul Aziz, H., 2022. Kinetic study of the anaerobic digestion of recycled paper mill effluent (RPME) by using a novel modified anaerobic hybrid Baffled (MAHB) reactor. Water 14, 390. https://doi.org/10.3390/w14030390.

Hoh, D., Watson, S., Kan, E., 2016. Algal biofilm reactors for integrated wastewater treatment and biofuel production: a review. Chem. Eng. J. 287, 466–473. https://doi.org/10.1016/j.cej.2015.11.062.

Hou, T., Zhao, J., Lei, Z., Shimizu, K., Zhang, Z., 2021. Enhanced energy recovery via separate hydrogen and methane production from two-stage anaerobic digestion of food waste with nanobubble water supplementation. Sci. Total Environ. 761, 143234 https://doi.org/10.1016/j.scitotenv.2020.143234.

Hou, Y.-P., Peng, D.-C., Zhang, A.-L., Wang, X.-B., Pei, L.-Y., 2022. Sludge granulation in ASBR: reactor performance, sludge physiochemical properties evolution and the unique clustered structure of the granular sludge. J. Water Process Eng. 49, 102948 https://doi.org/10.1016/j.jwpe.2022.102948.

Hu, Y., Cai, X., Du, R., Yang, Y., Rong, C., Qin, Y., Li, Y.-Y., 2022. A review on anaerobic membrane bioreactors for enhanced valorization of urban organic wastes: Achievements, limitations, energy balance and future perspectives. Sci. Total Environ. 820, 153284 https://doi.org/10.1016/j.scitotenv.2022.153284.

Huang, D.-Q., Fu, J.-J., Li, Z.-Y., Fan, N.-S., Jin, R.-C., 2022. Inhibition of wastewater pollutants on the anammox process: a review. Sci. Total Environ. 803, 150009 https://doi.org/10.1016/j.scitotenv.2021.150009.

Jallouli, W., Keskes, S., Guidara, W., Rezgui, F., Sayadi, S., Tounsi, S., 2021. Acidic pretreatment as a chemical approach for enhanced Photorhabdus temperata bioinsecticide production from industrial wastewater. J. Environ. Manage. 278, 111476 https://doi.org/10.1016/j.jenvman.2020.111476.

Jiang, H., Peng, Y., Li, X., Zhang, F., Wang, Z., Ren, S., 2020. Advanced nitrogen removal from mature landfill leachate via partial nitrification-Anammox biofilm reactor (PNABR) driven by high dissolved oxygen (DO): Protection mechanism of aerobic biofilm. Bioresour. Technol. 306, 123119 https://doi.org/10.1016/j.biortech.2020.123119.

Jiang, Y., McAdam, E., Zhang, Y., Heaven, S., Banks, C., Longhurst, P., 2019. Ammonia inhibition and toxicity in anaerobic digestion: a critical review. J. Water Process Eng. 32, 100899 https://doi.org/10.1016/j.jwpe.2019.100899.

Kamali, M., Appels, L., Kwon, E.E., Aminabhavi, T.M., Dewil, R., 2021. Biochar in water and wastewater treatment - a sustainability assessment. Chem. Eng. J. 420, 129946 https://doi.org/10.1016/j.cej.2021.129946.

Kanai, M., Ferre, V., Wakahara, S., Yamamoto, T., Moro, M., 2010. A novel combination of methane fermentation and MBR - kubota Submerged Anaerobic Membrane Bioreactor process. Desalination 250, 964–967. https://doi.org/10.1016/j.desal.2009.09.082.

Karki, R., Chuenchart, W., Surendra, K.C., Shrestha, S., Raskin, L., Sung, S., Hashimoto, A., Kumar Khanal, S., 2021. Anaerobic co-digestion: current status and perspectives. Bioresour. Technol. 330, 125001 https://doi.org/10.1016/j.biortech.2021.125001.

Kim, H.H., Saha, S., Hwang, J.-H., Hosen, MdA., Ahn, Y.-T., Park, Y.-K., Khan, M.A., Jeon, B.-H., 2022. Integrative biohydrogen- and biomethane-producing bioprocesses for comprehensive production of biohythane. Bioresour. Technol. 365, 128145 https://doi.org/10.1016/j.biortech.2022.128145.

Kim, J., Kang, C.-M., 2015. Increased anaerobic production of methane by co-digestion of sludge with microalgal biomass and food waste leachate. Bioresour. Technol. 189, 409–412. https://doi.org/10.1016/j.biortech.2015.04.028.

Kim, J.S., Lee, Y.Y., Kim, T.H., 2016. A review on alkaline pretreatment technology for bioconversion of lignocellulosic biomass. Bioresour. Technol. 199, 42–48. https://doi.org/10.1016/j.biortech.2015.08.085.

Kong, Z., Hao, T., Chen, H., Xue, Y., Li, D., Pan, Y., Li, Y., Li, Y.-Y., Huang, Y., 2023. Anaerobic membrane bioreactor for carbon-neutral treatment of industrial wastewater containing N, N-dimethylformamide: evaluation of electricity, bio-energy production and carbon emission. Environ. Res. 216, 114615 https://doi.org/10.1016/j.envres.2022.114615.

Kong, Z., Li, L., Kurihara, R., Kubota, K., Li, Y.-Y., 2018. Anaerobic treatment of N, N-dimethylformamide-containing wastewater by co-culturing two sources of inoculum. Water Res. 139, 228–239. https://doi.org/10.1016/j.watres.2018.03.078.

Kong, Z., Li, L., Kurihara, R., Zhang, T., Li, Y.-Y., 2019a. Anaerobic treatment of N,N-dimethylformamide-containing high-strength wastewater by submerged anaerobic membrane bioreactor with a co-cultured inoculum. Sci. Total Environ. 663, 696–708. https://doi.org/10.1016/j.scitotenv.2019.01.358.

Kong, Z., Li, L., Wu, J., Wang, T., Rong, C., Luo, Z., Pan, Y., Li, D., Li, Y., Huang, Y., Li, Y.-Y., 2021. Evaluation of bio-energy recovery from the anaerobic treatment of municipal wastewater by a pilot-scale submerged anaerobic membrane bioreactor (AnMBR) at ambient temperature. Bioresour. Technol. 339, 125551 https://doi.org/10.1016/j.biortech.2021.125551.

Kong, Z., Li, L., Xue, Y., Yang, M., Li, Y.-Y., 2019b. Challenges and prospects for the anaerobic treatment of chemical-industrial organic wastewater: a review. J. Clean. Prod. 231, 913–927. https://doi.org/10.1016/j.jclepro.2019.05.233.

Kong, Z., Xue, Y., Hao, T., Wu, J., Zhang, Y., Rong, C., Chen, H., Song, L., Li, D., Pan, Y., Li, Y.-Y., 2022a. Carbon-neutral treatment of N, N-dimethylformamide-containing wastewater by up-flow anaerobic sludge blanket: CO2 reduction and bio-energy cleaner production. J. Clean. Prod. 380, 134880 https://doi.org/10.1016/j.jclepro.2022.134880.

Kong, Z., Xue, Y., Hao, T., Zhang, Y., Wu, J., Chen, H., Song, L., Rong, C., Li, D., Pan, Y., Li, Y., Li, Y.-Y., 2022b. Carbon-neutral treatment of N, N-dimethylformamide-containing industrial wastewater by anaerobic membrane bioreactor (AnMBR): bio-energy recovery and CO2 emission reduction. Bioresour. Technol. 358, 127396 https://doi.org/10.1016/j.biortech.2022.127396.

Kong, Z., Xue, Y., Hao, T., Zhang, Y., Wu, J., Chen, H., Song, L., Rong, C., Li, D., Pan, Y., Li, Y., Li, Y.-Y., 2022c. Carbon-neutral treatment of N, N-dimethylformamide-containing industrial wastewater by anaerobic membrane bioreactor (AnMBR): bio-energy recovery and CO2 emission reduction. Bioresour. Technol. 358, 127396 https://doi.org/10.1016/j.biortech.2022.127396.

Kong, Z., Xue, Y., Zhang, Y., Hao, T., Chen, H., Sun, J., Pan, Y., Li, D., Li, Y., Huang, Y., 2022d. Insights into the carbon neutrality for the treatment process engineering of municipal wastewater by anaerobic membrane bioreactor integrated with partial nitritation-anammox: CO2 reduction and energy recovery. J. Water Process Eng. 49, 102996 https://doi.org/10.1016/j.jwpe.2022.102996.

Kougias, P.G., Kotsopoulos, T.A., Martzopoulos, G.G., 2014. Effect of feedstock composition and organic loading rate during the mesophilic co-digestion of olive mill wastewater and swine manure. Renew. Energy 69, 202–207. https://doi.org/10.1016/j.renene.2014.03.047.

Koul, Y., Devda, V., Varjani, S., Guo, W., Ngo, H.H., Taherzadeh, M.J., Chang, J.-S., Wong, J.W.C., Bilal, M., Kim, S.-H., Bui, X.-T., Parra-Saldivar, R., 2022. Microbial electrolysis: a promising approach for treatment and resource recovery from industrial wastewater. Bioengineered 13, 8115–8134. https://doi.org/10.1080/21655979.2022.2051842.

Krayzelova, L., Bartacek, J., Díaz, I., Jeison, D., Volcke, E.I.P., Jenicek, P., 2015. Microaeration for hydrogen sulfide removal during anaerobic treatment: a review. Rev. Environ. Sci. Biotechnol. 14, 703–725. https://doi.org/10.1007/s11157-015-9386-2.

Kunatsa, T., Xia, X., 2022. A review on anaerobic digestion with focus on the role of biomass co-digestion, modelling and optimisation on biogas production and enhancement. Bioresour. Technol. 344, 126311 https://doi.org/10.1016/j.biortech.2021.126311.

Kuypers, M.M.M., Marchant, H.K., Kartal, B., 2018. The microbial nitrogen-cycling network. Nat. Rev. Microbiol. 16, 263–276. https://doi.org/10.1038/nrmicro.2018.9.

Lei, Z., Ma, Y., Wang, J., Wang, X.C., Li, Q., Chen, R., 2021. Biochar addition supports high digestion performance and low membrane fouling rate in an anaerobic membrane bioreactor under low temperatures. Bioresour. Technol. 330, 124966 https://doi.org/10.1016/j.biortech.2021.124966.

Leite, W.R.M., Gottardo, M., Pavan, P., Belli Filho, P., Bolzonella, D., 2016. Performance and energy aspects of single and two phase thermophilic anaerobic digestion of waste activated sludge. Renew. Energy 86, 1324–1331. https://doi.org/10.1016/j.renene.2015.09.069.

Li, D., Tan, Y., Zhou, Y., Pathak, S., Sendjaja, A.Y., Abdul Majid, M., Chowdhury, P., Ng, W.J., 2015. Comparative study of low-energy ultrasonic and alkaline treatment on biosludge from secondary industrial wastewater treatment. Environ. Technol. 36, 2239–2248. https://doi.org/10.1080/09593330.2015.1025103.

Li, J., Lu, S., Wu, S., Zhang, W., Hua, M., Pan, B., 2022. The breakdown of protein hydrogen bonding networks facilitates biotransformation of protein wastewaters during anaerobic digestion methanogenesis: focus on protein structure and conformation. Environ. Res. 208, 112735 https://doi.org/10.1016/j.envres.2022.112735.

Li, L., Qin, Y., Kong, Z., Wu, J., Kubota, K., Li, Y.-Y., 2019. Characterization of microbial community and main functional groups of prokaryotes in thermophilic anaerobic co-digestion of food waste and paper waste. Sci. Total Environ. 652, 709–717. https://doi.org/10.1016/j.scitotenv.2018.10.292.

Li, L., Wang, X., Miao, J., Abulimiti, A., Jing, X., Ren, N., 2022. Carbon neutrality of wastewater treatment - a systematic concept beyond the plant boundary. Environ. Sci. Ecotechnology 11, 100180. https://doi.org/10.1016/j.ese.2022.100180.

Li, Q., Liu, Y., Yang, X., Zhang, J., Lu, B., Chen, R., 2020. Kinetic and thermodynamic effects of temperature on methanogenic degradation of acetate, propionate, butyrate and valerate. Chem. Eng. J. 396, 125366 https://doi.org/10.1016/j.cej.2020.125366.

Li, X., Yuan, Y., Huang, Y., Bi, Z., 2019. Simultaneous removal of ammonia and nitrate by coupled S0-driven autotrophic denitrification and Anammox process in fluorine-containing semiconductor wastewater. Sci. Total Environ. 661, 235–242. https://doi.org/10.1016/j.scitotenv.2019.01.164.

Li, Y., Chen, Y., Wu, J., 2019. Enhancement of methane production in anaerobic digestion process: a review. Appl. Energy 240, 120–137. https://doi.org/10.1016/j.apenergy.2019.01.243.

Liang, J., Mai, W., Tang, J., Wei, Y., 2019. Highly effective treatment of petrochemical wastewater by a super-sized industrial scale plant with expanded granular sludge bed bioreactor and aerobic activated sludge. Chem. Eng. J. 360, 15–23. https://doi.org/10.1016/j.cej.2018.11.167.

Liang, Y.-C., Daverey, A., Huang, Y.-T., Sung, S., Lin, J.-G., 2016. Treatment of semiconductor wastewater using single-stage partial nitrification and anammox in a pilot-scale reactor. J. Taiwan Inst. Chem. Eng. 63, 236–242. https://doi.org/10.1016/j.jtice.2016.02.036.

Lim, S.J., Kim, T.-H., 2014. Applicability and trends of anaerobic granular sludge treatment processes. Biomass Bioenergy 60, 189–202. https://doi.org/10.1016/j.biombioe.2013.11.011.

Liu, S., 2023. A comprehensive literature mining and analysis of nitrous oxide emissions from different innovative mainstream anammox-based biological nitrogen removal processes. Sci. Total Environ. 902, 166295. https://doi.org/10.1016/j.scitotenv.2023.166295.

Liu, Y., Xiao, Q., Jia, Z., Wang, C., Ye, X., Du, J., Kong, X., Xi, Y., 2021. Relieving ammonia nitrogen inhibition in high concentration anaerobic digestion of rural organic household waste by Prussian blue analogue nanoparticles addition. Bioresour. Technol. 330, 124979 https://doi.org/10.1016/j.biortech.2021.124979.

Loow, Y.-L., Wu, T.Y., Jahim, Md, Mohammad, A.W., Teoh, W.H., 2016. Typical conversion of lignocellulosic biomass into reducing sugars using dilute acid hydrolysis and alkaline pretreatment. Cellulose 23, 1491–1520. https://doi.org/10.1007/s10570-016-0936-8.

Lorenci Woiciechowski, A., Dalmas Neto, C.J., Porto de Souza Vandenberghe, L., de Carvalho Neto, D.P., Novak Sydney, A.C., Letti, L.A.J., Karp, S.G., Zevallos Torres, L. A., Soccol, C.R., 2020. Lignocellulosic biomass: acid and alkaline pretreatments and their effects on biomass recalcitrance – conventional processing and recent advances. Bioresour. Technol. 304, 122848 https://doi.org/10.1016/j.biortech.2020.122848.

Lu, D., Xiao, K., Chen, Y., Soh, Y.N.A., Zhou, Y., 2018. Transformation of dissolved organic matters produced from alkaline-ultrasonic sludge pretreatment in anaerobic digestion: from macro to micro. Water Res. 142, 138–146. https://doi.org/10.1016/j.watres.2018.05.044.

Lü, F., Luo, C., Shao, L., He, P., 2016. Biochar alleviates combined stress of ammonium and acids by firstly enriching Methanosaeta and then Methanosarcina. Water Res. 90, 34–43. https://doi.org/10.1016/j.watres.2015.12.029.

Lu, X., Zheng, C., Zhen, G., Tan, Y., Zhou, Y., Zhang, Z., Niu, C., Li, W., Kudisi, D., Wang, Y., Li, Y.-Y., 2021. Roles of colloidal particles and soluble biopolymers in long-term performance and fouling behaviors of submerged anaerobic membrane bioreactor treating methanolic wastewater. J. Clean. Prod. 290, 125816 https://doi.org/10.1016/j.jclepro.2021.125816.

Ma, B., Wang, S., Cao, S., Miao, Y., Jia, F., Du, R., Peng, Y., 2016. Biological nitrogen removal from sewage via anammox: recent advances. Bioresour. Technol. 200, 981–990. https://doi.org/10.1016/j.biortech.2015.10.074.

Macintosh, C., Astals, S., Sembera, C., Ertl, A., Drewes, J.E., Jensen, P.D., Koch, K., 2019. Successful strategies for increasing energy self-sufficiency at Grüneck wastewater treatment plant in Germany by food waste co-digestion and improved aeration. Appl. Energy 242, 797–808. https://doi.org/10.1016/j.apenergy.2019.03.126.

Mai, W., Chen, J., Liu, H., Liang, J., Tang, J., Wei, Y., 2021. Advances in studies on microbiota involved in nitrogen removal processes and their applications in wastewater treatment. Front. Microbiol. 12, 746293 https://doi.org/10.3389/fmicb.2021.746293.

Maleki, E., 2020. Psychrophilic anaerobic membrane bioreactor (AnMBR) for treating malting plant wastewater and energy recovery. J. Water Process Eng. 34, 101174 https://doi.org/10.1016/j.jwpe.2020.101174.

Mamais, D., Noutsopoulos, C., Dimopoulou, A., Stasinakis, A., Lekkas, T.D., 2015. Wastewater treatment process impact on energy savings and greenhouse gas emissions. Water Sci. Technol. 71, 303–308. https://doi.org/10.2166/wst.2014.521.

Meena, M., Yadav, G., Sonigra, P., Shah, M.P., 2022. A comprehensive review on application of bioreactor for industrial wastewater treatment. Lett. Appl. Microbiol. 74, 131–158. https://doi.org/10.1111/lam.13557.

Meng, L.-W., Li, X.-K., Wang, K., Ma, K.-L., Zhang, J., 2016. Pre-treating amoxicillin contained wastewater with an anaerobic expanded granular sludge bed (EGSB). Desalination Water Treat. 57, 16008–16014. https://doi.org/10.1080/19443994.2015.1075424.

Mishra, P., Thakur, S., Mahapatra, D.M., Wahid, Z.A., Liu, H., Singh, L., 2018. Impacts of nano-metal oxides on hydrogen production in anaerobic digestion of palm oil mill effluent – a novel approach. Int. J. Hydrogen Energy 43, 2666–2676. https://doi.org/10.1016/j.ijhydene.2017.12.108.

Mizuta, K., Shimada, M., 2010. Benchmarking energy consumption in municipal wastewater treatment plants in Japan. Water Sci. Technol. 62, 2256–2262. https://doi.org/10.2166/wst.2010.510.

Mohd Udaiyappan, A.F., Abu Hasan, H., Takriff, M.S., Sheikh Abdullah, S.R., 2017. A review of the potentials, challenges and current status of microalgae biomass applications in industrial wastewater treatment. J. Water Process Eng. 20, 8–21. https://doi.org/10.1016/j.jwpe.2017.09.006.

Moñino, P., Aguado, D., Barat, R., Jiménez, E., Giménez, J.B., Seco, A., Ferrer, J., 2017. A new strategy to maximize organic matter valorization in municipalities: combination of urban wastewater with kitchen food waste and its treatment with AnMBR technology. Waste Manag. 62, 274–289. https://doi.org/10.1016/j.wasman.2017.02.006.

Mulu, E., M'Arimi, M.M., Ramkat, R.C., Mecha, A.C., 2021. Potential of wood ash in purification of biogas. Energy Sustain. Dev. 65, 45–52. https://doi.org/10.1016/j.esd.2021.09.009.

Nabi, M., Liang, H., Zhou, Q., Cao, J., Gao, D., 2023. In-situ membrane fouling control and performance improvement by adding materials in anaerobic membrane bioreactor: a review. Sci. Total Environ. 865, 161262 https://doi.org/10.1016/j.scitotenv.2022.161262.

Nguyen, D., Khanal, S.K., 2018. A little breath of fresh air into an anaerobic system: how microaeration facilitates anaerobic digestion process. Biotechnol. Adv. 36, 1971–1983. https://doi.org/10.1016/j.biotechadv.2018.08.007.

Nguyen, D., Wu, Z., Shrestha, S., Lee, P.-H., Raskin, L., Khanal, S.K., 2019. Intermittent micro-aeration: new strategy to control volatile fatty acid accumulation in high organic loading anaerobic digestion. Water Res. 166, 115080 https://doi.org/10.1016/j.watres.2019.115080.

Ni, Z., Zhou, L., Lin, Z., Kuang, B., Zhu, G., Jia, J., Wang, T., 2023. Iron-modified biochar boosts anaerobic digestion of sulfamethoxazole pharmaceutical wastewater: performance and microbial mechanism. J. Hazard Mater. 452, 131314 https://doi.org/10.1016/j.jhazmat.2023.131314.

Nidheesh, P.V., Kumar, A., Syam Babu, D., Scaria, J., Suresh Kumar, M., 2020. Treatment of mixed industrial wastewater by electrocoagulation and indirect electrochemical oxidation. Chemosphere 251, 126437. https://doi.org/10.1016/j.chemosphere.2020.126437.

Oduor, W.W., Wandera, S.M., Murunga, S.I., Raude, J.M., 2022. Enhancement of anaerobic digestion by co-digesting food waste and water hyacinth in improving treatment of organic waste and bio-methane recovery. Heliyon 8, e10580. https://doi.org/10.1016/j.heliyon.2022.e10580.

Pan, W., Ouyang, H., Tan, X., Yan, S., Zhang, R., Deng, R., Gu, L., He, Q., 2023. Effects of biochar addition towards the treatment of blackwater in anaerobic dynamic membrane bioreactor (AnDMBR): comparison among room temperature, mesophilic and thermophilic conditions. Bioresour. Technol. 374, 128776 https://doi.org/10.1016/j.biortech.2023.128776.

Pan, X., Lv, N., Cai, G., Zhou, M., Wang, R., Li, C., Ning, J., Li, J., Li, Y., Ye, Z., Zhu, G., 2021. Carbon- and metal-based mediators modulate anaerobic methanogenesis and phenol removal: focusing on stimulatory and inhibitory mechanism. J. Hazard Mater. 420, 126615 https://doi.org/10.1016/j.jhazmat.2021.126615.

Paranjpe, A., Saxena, S., Jain, P., 2023. A review on performance improvement of anaerobic digestion using Co-digestion of food waste and sewage sludge. J. Environ. Manage. 338, 117733 https://doi.org/10.1016/j.jenvman.2023.117733.

Pretel, R., Robles, A., Ruano, M.V., Seco, A., Ferrer, J., 2014. The operating cost of an anaerobic membrane bioreactor (AnMBR) treating sulphate-rich urban wastewater. Sep. Purif. Technol. 126, 30–38. https://doi.org/10.1016/j.seppur.2014.02.013.

Promnuan, K., Higuchi, T., Imai, T., Kongjan, P., Reungsang, A., O-Thong, S., 2020. Simultaneous biohythane production and sulfate removal from rubber sheet wastewater by two-stage anaerobic digestion. Int. J. Hydrogen Energy 45, 263–274. https://doi.org/10.1016/j.ijhydene.2019.10.237.

Pyae, H.A., Boontian, N., Yingchon, U., Piasai, C., 2018. Effect of acid-alkaline and thermal pre-treatment to cassava pulp feed for batch reactors in optimization of bio-methane yield. Appl. Environ. Res. 51–60. https://doi.org/10.35762/AER.2018.40.2.5.

Rashvanlou, R.B., Farzadkia, M., Rezaee, A., Gholami, M., Kermani, M., Pasalari, H., 2021. The influence of combined low-strength ultrasonics and micro-aerobic pretreatment process on methane generation and sludge digestion: lipase enzyme, microbial activation, and energy yield. Ultrason. Sonochem. 73, 105531 https://doi.org/10.1016/j.ultsonch.2021.105531.

Ren, Y., Yu, M., Wu, C., Wang, Q., Gao, M., Huang, Q., Liu, Y., 2018. A comprehensive review on food waste anaerobic digestion: research updates and tendencies. Bioresour. Technol. 247, 1069–1076. https://doi.org/10.1016/j.biortech.2017.09.109.

Rong, C., Luo, Z., Wang, T., Guo, Y., Kong, Z., Wu, J., Ji, J., Qin, Y., Hanaoka, T., Sakemi, S., Ito, M., Kobayashi, S., Kobayashi, M., Li, Y.-Y., 2021. Chemical oxygen demand and nitrogen transformation in a large pilot-scale plant with a combined submerged anaerobic membrane bioreactor and one-stage partial nitritation-anammox for treating mainstream wastewater at 25 °C. Bioresour. Technol. 341, 125840 https://doi.org/10.1016/j.biortech.2021.125840.

Šabić Runjavec, M., Vuković Domanovac, M., Meštrović, E., 2022. Removal of organic pollutants from real pharmaceutical industrial wastewater with environmentally friendly processes. Chem. Pap. 76, 1423–1431. https://doi.org/10.1007/s11696-021-01919-x.

Sanchis-Perucho, P., Robles, Á., Durán, F., Rogalla, F., Ferrer, J., Seco, A., 2021. Widening the applicability of AnMBR for urban wastewater treatment through PDMS membranes for dissolved methane capture: effect of temperature and hydrodynamics. J. Environ. Manage. 287, 112344 https://doi.org/10.1016/j.jenvman.2021.112344.

Saravanan, A., Deivayanai, V.C., Kumar, P.S., Rangasamy, G., Hemavathy, R.V., Harshana, T., Gayathri, N., Alagumalai, K., 2022. A detailed review on advanced oxidation process in treatment of wastewater: mechanism, challenges and future outlook. Chemosphere 308, 136524. https://doi.org/10.1016/j.chemosphere.2022.136524.

Saravanan, A., Senthil Kumar, P., Rangasamy, G., Hariharan, R., Hemavathy, R.V., Deepika, P.D., Anand, K., Karthika, S., 2023. Strategies for enhancing the efficacy of anaerobic digestion of food industry wastewater: an insight into bioreactor types, challenges, and future scope. Chemosphere 310, 136856. https://doi.org/10.1016/j.chemosphere.2022.136856.

Shabir, S., Ilyas, N., Mashwani, Z.-R., Ahmad, M.S., Al-Ansari, M.M., Al-Humaid, L., Reddy, M.S., 2022. Designing of pretreatment filter technique for reduction of phenolic constituents from olive-mill wastewater and testing its impact on wheat germination. Chemosphere 299, 134438. https://doi.org/10.1016/j.chemosphere.2022.134438.

Shah, A., Shah, M., 2020. Characterisation and bioremediation of wastewater: a review exploring bioremediation as a sustainable technique for pharmaceutical wastewater. Groundw. Sustain. Dev. 11, 100383 https://doi.org/10.1016/j.gsd.2020.100383.

Sharma, P., Tripathi, S., Vadakedath, N., Chandra, R., 2021. In-situ toxicity assessment of pulp and paper industry wastewater on Trigonella foenum-graecum L: potential source of cytotoxicity and chromosomal damage. Environ. Technol. Innov. 21, 101251 https://doi.org/10.1016/j.eti.2020.101251.

Sharma, R., Malaviya, P., 2022. Constructed wetlands for textile wastewater remediation: a review on concept, pollutant removal mechanisms, and integrated technologies for efficiency enhancement. Chemosphere 290, 133358. https://doi.org/10.1016/j.chemosphere.2021.133358.

Shen, M.-Y., Huang, Z.-H., Kuo, Y.-T., Hsu, J., Yang, H.-W., Peng, C.-Y., Chu, C.-Y., 2022. Improvement of gaseous bioenergy production from spent coffee grounds Co-digestion with pulp wastewater by physical/chemical pretreatments. Int. J. Hydrogen Energy 47, 40664–40671. https://doi.org/10.1016/j.ijhydene.2022.10.041.

Sheng, L., Lei, Z., Dzakpasu, M., Li, Y.-Y., Li, Q., Chen, R., 2020a. Application of the anammox-based process for nitrogen removal from anaerobic digestion effluent: a review of treatment performance, biochemical reactions, and impact factors. J. Water Process Eng. 38, 101595 https://doi.org/10.1016/j.jwpe.2020.101595.

Sheng, L., Lei, Z., Dzakpasu, M., Li, Y.-Y., Li, Q., Chen, R., 2020b. Application of the anammox-based process for nitrogen removal from anaerobic digestion effluent: a review of treatment performance, biochemical reactions, and impact factors. J. Water Process Eng. 38, 101595 https://doi.org/10.1016/j.jwpe.2020.101595.

Shi, J., Wan, N., Li, L., Li, Z., Han, H., 2022. Review on treatment technologies of coal gasification wastewater in China. J. Clean. Prod. 333, 130166 https://doi.org/10.1016/j.jclepro.2021.130166.

Siatou, A., Manali, A., Gikas, P., 2020. Energy consumption and internal distribution in activated sludge wastewater treatment plants of Greece. Water 12, 1204. https://doi.org/10.3390/w12041204.

Siddique, MdN.I., Wahid, Z.Ab, 2018. Achievements and perspectives of anaerobic co-digestion: a review. J. Clean. Prod. 194, 359–371. https://doi.org/10.1016/j.jclepro.2018.05.155.

Silva, C., Rosa, M.J., 2022. A comprehensive derivation and application of Reference values for benchmarking the energy performance of activated sludge wastewater treatment. Water 14, 1620. https://doi.org/10.3390/w14101620.

Silveira, N.C., Oliveira, G.H.D., Damianovic, M.H.R.Z., Foresti, E., 2021. Two-stage partial nitrification-Anammox process for nitrogen removal from slaughterhouse wastewater: evaluation of the nitrogen loading rate and microbial community analysis. J. Environ. Manage. 296, 113214 https://doi.org/10.1016/j.jenvman.2021.113214.

Singh, A., Srivastava, A., Saidulu, D., Gupta, A.K., 2022. Advancements of sequencing batch reactor for industrial wastewater treatment: major focus on modifications, critical operational parameters, and future perspectives. J. Environ. Manage. 317, 115305 https://doi.org/10.1016/j.jenvman.2022.115305.

Smith, A.L., Stadler, L.B., Cao, L., Love, N.G., Raskin, L., Skerlos, S.J., 2014. Navigating wastewater energy recovery strategies: a life cycle comparison of anaerobic membrane bioreactor and conventional treatment systems with anaerobic digestion. Environ. Sci. Technol. 48, 5972–5981. https://doi.org/10.1021/es5006169.

Sohn, W., Guo, W., Ngo, H.H., Deng, L., Cheng, D., 2021. Powdered activated carbon addition for fouling control in anaerobic membrane bioreactor. Bioresour. Technol. Rep. 15, 100721 https://doi.org/10.1016/j.biteb.2021.100721.

Sołowski, G., Konkol, I., Cenian, A., 2020. Production of hydrogen and methane from lignocellulose waste by fermentation. A review of chemical pretreatment for enhancing the efficiency of the digestion process. J. Clean. Prod. 267, 121721 https://doi.org/10.1016/j.jclepro.2020.121721.

Song, Y.-C., Kim, M., Shon, H., Jegatheesan, V., Kim, S., 2018. Modeling methane production in anaerobic forward osmosis bioreactor using a modified anaerobic digestion model No. 1. Bioresour. Technol. 264, 211–218. https://doi.org/10.1016/j.biortech.2018.04.125.

Sturm, B.S.M., Lamer, S.L., 2011. An energy evaluation of coupling nutrient removal from wastewater with algal biomass production. Appl. Energy 88, 3499–3506. https://doi.org/10.1016/j.apenergy.2010.12.056.

Subbarao, P.M.V., D' Silva, T.C., Adlak, K., Kumar, S., Chandra, R., Vijay, V.K., 2023. Anaerobic digestion as a sustainable technology for efficiently utilizing biomass in the context of carbon neutrality and circular economy. Environ. Res. 116286 https://doi.org/10.1016/j.envres.2023.116286.

Sun, P., Liu, C., Li, A., Ji, B., 2022. Using carbon dioxide-added microalgal-bacterial granular sludge for carbon-neutral municipal wastewater treatment under outdoor conditions: performance, granule characteristics and environmental sustainability. Sci. Total Environ. 848, 157657 https://doi.org/10.1016/j.scitotenv.2022.157657.

Sushma Kumari, M., Saroha, A.K., 2018. Performance of various catalysts on treatment of refractory pollutants in industrial wastewater by catalytic wet air oxidation: a review. J. Environ. Manage. 228, 169–188. https://doi.org/10.1016/j.jenvman.2018.09.003.

Tamang, M., Paul, K.K., 2022. Advances in treatment of coking wastewater – a state of art review. Water Sci. Technol. 85, 449–473. https://doi.org/10.2166/wst.2021.497.

Toczyłowska-Mamińska, R., Mamiński, M.Ł., 2022. Wastewater as a renewable energy source—utilisation of microbial fuel cell technology. Energies 15, 6928. https://doi.org/10.3390/en15196928.

Toumi, J., Miladi, B., Farhat, A., Nouira, S., Hamdi, M., Gtari, M., Bouallagui, H., 2015. Microbial ecology overview during anaerobic codigestion of dairy wastewater and cattle manure and use in agriculture of obtained bio-fertilisers. Bioresour. Technol. 198, 141–149. https://doi.org/10.1016/j.biortech.2015.09.004.

Trinh, H.P., Lee, S.-H., Jeong, G., Yoon, H., Park, H.-D., 2021. Recent developments of the mainstream anammox processes: challenges and opportunities. J. Environ. Chem. Eng. 9, 105583 https://doi.org/10.1016/j.jece.2021.105583.

Vaccari, M., Foladori, P., Nembrini, S., Vitali, F., 2018. Benchmarking of energy consumption in municipal wastewater treatment plants – a survey of over 200 plants in Italy. Water Sci. Technol. 77, 2242–2252. https://doi.org/10.2166/wst.2018.035.

Val Del Rio, A., Pichel, A., Fernandez-Gonzalez, N., Pedrouso, A., Fra-Vázquez, A., Morales, N., Mendez, R., Campos, J.L., Mosquera-Corral, A., 2018. Performance and microbial features of the partial nitritation-anammox process treating fish canning wastewater with variable salt concentrations. J. Environ. Manage. 208, 112–121. https://doi.org/10.1016/j.jenvman.2017.12.007.

Vítězová, M., Kohoutová, A., Vítěz, T., Hanišáková, N., Kushkevych, I., 2020. Methanogenic microorganisms in industrial wastewater anaerobic treatment. Processes 8, 1546. https://doi.org/10.3390/pr8121546.

Wang, C., Liu, Y., Gao, X., Chen, H., Xu, X., Zhu, L., 2018. Role of biochar in the granulation of anaerobic sludge and improvement of electron transfer characteristics. Bioresour. Technol. 268, 28–35. https://doi.org/10.1016/j.biortech.2018.07.116.

Wang, Chen, Liu, Y., Wang, Caiqin, Xing, B., Zhu, S., Huang, J., Xu, X., Zhu, L., 2021. Biochar facilitates rapid restoration of methanogenesis by enhancing direct interspecies electron transfer after high organic loading shock. Bioresour. Technol. 320, 124360 https://doi.org/10.1016/j.biortech.2020.124360.

Wang, D., Han, H., Han, Y., Li, K., Zhu, H., 2017. Enhanced treatment of Fischer-Tropsch (F-T) wastewater using the up-flow anaerobic sludge blanket coupled with bioelectrochemical system: effect of electric field. Bioresour. Technol. 232, 18–26. https://doi.org/10.1016/j.biortech.2017.02.010.

Wang, L., Hao, J., Wang, C., Li, Y., Yang, Q., 2022. Carbohydrate-to-protein ratio regulates hydrolysis and acidogenesis processes during volatile fatty acids production. Bioresour. Technol. 355, 127266 https://doi.org/10.1016/j.biortech.2022.127266.

Wang, S., Hu, J., He, S., Wang, J., 2022. Removal of ammonia and phenol from saline chemical wastewater by ionizing radiation: performance, mechanism and toxicity. J. Hazard Mater. 433, 128727 https://doi.org/10.1016/j.jhazmat.2022.128727.

Wang, W., Chang, J.-S., Lee, D.-J., 2022. Integrating anaerobic digestion with bioelectrochemical system for performance enhancement: a mini review. Bioresour. Technol. 345, 126519 https://doi.org/10.1016/j.biortech.2021.126519.

Wang, Z., Zhao, Y., Ge, Z., Zhang, H., Sun, S., 2016. Selection of microalgae for simultaneous biogas upgrading and biogas slurry nutrient reduction under various photoperiods: selection of microalgal strains for biogas upgrading. J. Chem. Technol. Biotechnol. 91, 1982–1989. https://doi.org/10.1002/jctb.4788.

Weralupitiya, C., Wanigatunge, R., Joseph, S., Athapattu, B.C.L., Lee, T.-H., Kumar Biswas, J., Ginige, M.P., Shiung Lam, S., Senthil Kumar, P., Vithanage, M., 2021. Anammox bacteria in treating ammonium rich wastewater: recent perspective and appraisal. Bioresour. Technol. 334, 125240 https://doi.org/10.1016/j.biortech.2021.125240.

Wu, C., Zhang, G., Zhang, P., Chang, C.-C., 2014. Disintegration of excess activated sludge with potassium permanganate: feasibility, mechanisms and parameter optimization. Chem. Eng. J. 240, 420–425. https://doi.org/10.1016/j.cej.2013.12.011.

Wu, C., Zhou, Y., Wang, P., Guo, S., 2015. Improving hydrolysis acidification by limited aeration in the pretreatment of petrochemical wastewater. Bioresour. Technol. 194, 256–262. https://doi.org/10.1016/j.biortech.2015.06.072.

Wu, J., Kong, Z., Luo, Z., Qin, Y., Rong, C., Wang, T., Hanaoka, T., Sakemi, S., Ito, M., Kobayashi, S., Kobayashi, M., Xu, K.-Q., Kobayashi, T., Kubota, K., Li, Y.-Y., 2021. A successful start-up of an anaerobic membrane bioreactor (AnMBR) coupled mainstream partial nitritation-anammox (PN/A) system: a pilot-scale study on in-situ NOB elimination, AnAOB growth kinetics, and mainstream treatment performance. Water Res. 207, 117783 https://doi.org/10.1016/j.watres.2021.117783.

Wu, J., Li, H., Wang, K., Zuo, J., 2019. Performance and microbial community of an expanded granular sludge bed reactor in the treatment of cephalosporin wastewater. Bioresour. Technol. 275, 94–100. https://doi.org/10.1016/j.biortech.2018.12.003.

Wu, J., Niu, Q., Li, L., Hu, Y., Mribet, C., Hojo, T., Li, Y.-Y., 2018. A gradual change between methanogenesis and sulfidogenesis during a long-term UASB treatment of sulfate-rich chemical wastewater. Sci. Total Environ. 636, 168–176. https://doi.org/10.1016/j.scitotenv.2018.04.172.

Wu, P., Chen, J., Garlapati, V.K., Zhang, Xingxing, Wani Victor Jenario, F., Li, X., Liu, W., Chen, C., Aminabhavi, T.M., Zhang, Xiaonong, 2022. Novel insights into Anammox-based processes: a critical review. Chem. Eng. J. 444, 136534 https://doi.org/10.1016/j.cej.2022.136534.

Xiao, R., Ni, B.-J., Liu, S., Lu, H., 2021. Impacts of organics on the microbial ecology of wastewater anammox processes: recent advances and meta-analysis. Water Res. 191, 116817 https://doi.org/10.1016/j.watres.2021.116817.

Xie, J., Zou, X., Chang, Y., Liu, H., Cui, M.-H., Zhang, T.C., Xi, J., Chen, C., 2023. A feasibility investigation of a pilot-scale bioelectrochemical coupled anaerobic digestion system with centric electrode module for real membrane manufacturing wastewater treatment. Bioresour. Technol. 368, 128371 https://doi.org/10.1016/j.biortech.2022.128371.

Xie, Jiawei, Chang, Y., Xie, Junxiang, Adams, M., Zhao, D., Chen, C., Ma, J., Zhu, G., Zhang, T.C., 2021. Insights into the mechanism, performance and electrode modification of BES-AD combined systems for refractory wastewater treatment: a review. J. Water Process Eng. 40, 101895 https://doi.org/10.1016/j.jwpe.2020.101895.

Xie, Q., Zhang, Y., Zhuang, H., Liang, R., Zhan, X., Zhao, Y., Shan, S., Wu, H., 2022. Enhanced degradation of high-concentrated 2,4,6 trichlorophenol-containing wastewater by micron-grade high iron-containing fly ash/loofah sponge biochar: metagenomic functional succession. J. Water Process Eng. 47, 102837 https://doi.org/10.1016/j.jwpe.2022.102837.

Xu, R., Zhang, K., Liu, P., Khan, A., Xiong, J., Tian, F., Li, X., 2018. A critical review on the interaction of substrate nutrient balance and microbial community structure and function in anaerobic co-digestion. Bioresour. Technol. 247, 1119–1127. https://doi.org/10.1016/j.biortech.2017.09.095.

Xu, Y., Liu, R., Yang, D., Dai, X., 2021. Sludge treatment and resource recovery towards carbon neutrality in China: current status and future perspective. Blue-Green Syst. 3, 119–127. https://doi.org/10.2166/bgs.2021.115.

Yadav, G., Mishra, A., Ghosh, P., Sindhu, R., Vinayak, V., Pugazhendhi, A., 2021. Technical, economic and environmental feasibility of resource recovery technologies from wastewater. Sci. Total Environ. 796, 149022 https://doi.org/10.1016/j.scitotenv.2021.149022.

Yan, M., Fotidis, I.A., Tian, H., Khoshnevisan, B., Treu, L., Tsapekos, P., Angelidaki, I., 2019. Acclimatization contributes to stable anaerobic digestion of organic fraction of municipal solid waste under extreme ammonia levels: focusing on microbial community dynamics. Bioresour. Technol. 286, 121376 https://doi.org/10.1016/j.biortech.2019.121376.

Yang, D., Chen, Q., Liu, R., Song, L., Zhang, Y., Dai, X., 2022. Ammonia recovery from anaerobic digestate: state of the art, challenges and prospects. Bioresour. Technol. 363, 127957 https://doi.org/10.1016/j.biortech.2022.127957.

Yang, L., Liu, Y., Li, C., Liu, Z., Liu, X., Wei, C., Yang, Z., Zhang, A., 2022. Biodegradation time series characteristics and metabolic fate of different aromatic compounds in the biochemical treatment process of coal chemical wastewater. Bioresour. Technol. 361, 127688 https://doi.org/10.1016/j.biortech.2022.127688.

Yang, L., Zeng, S., Chen, J., He, M., Yang, W., 2010. Operational energy performance assessment system of municipal wastewater treatment plants. Water Sci. Technol. 62, 1361–1370. https://doi.org/10.2166/wst.2010.394.

Yang, W., Li, S., Qv, M., Dai, D., Liu, D., Wang, W., Tang, C., Zhu, L., 2022. Microalgal cultivation for the upgraded biogas by removing CO2, coupled with the treatment of slurry from anaerobic digestion: a review. Bioresour. Technol. 364, 128118 https://doi.org/10.1016/j.biortech.2022.128118.

Yang, X., Wei, J., Ye, G., Zhao, Y., Li, Z., Qiu, G., Li, F., Wei, C., 2020. The correlations among wastewater internal energy, energy consumption and energy recovery/production potentials in wastewater treatment plant: an assessment of the energy balance. Sci. Total Environ. 714, 136655 https://doi.org/10.1016/j.scitotenv.2020.136655.

Yao, H., Zhao, X., Fan, L., Jia, F., Chen, Y., Cai, W., Guo, J., 2022. Pilot-scale demonstration of one-stage partial nitritation/anammox process to treat wastewater from a coal to ethylene glycol (CtEG) plant. Environ. Res. 208, 112540 https://doi.org/10.1016/j.envres.2021.112540.

Ye, M., Li, Q., Li, Y.-Y., 2023. Evaluation of anaerobic membrane bioreactor treating dairy processing wastewater: elemental flow, bioenergy production and reduction of CO2 emission. Bioresour. Technol. 385, 129342 https://doi.org/10.1016/j.biortech.2023.129342.

Yin, Y., Zhang, Q., Peng, H., 2023. Retrospect and prospect of aerobic biodegradation of aniline: overcome existing bottlenecks and follow future trends. J. Environ. Manage. 330, 117133 https://doi.org/10.1016/j.jenvman.2022.117133.

Yuan, H., Zhu, N., 2016. Progress in inhibition mechanisms and process control of intermediates and by-products in sewage sludge anaerobic digestion. Renew. Sustain. Energy Rev. 58, 429–438. https://doi.org/10.1016/j.rser.2015.12.261.

Yuan, Y., Cheng, H., Chen, F., Zhang, Y., Xu, X., Huang, C., Chen, C., Liu, W., Ding, C., Li, Z., Chen, T., Wang, A., 2020. Enhanced methane production by alleviating sulfide inhibition with a microbial electrolysis coupled anaerobic digestion reactor. Environ. Int. 136, 105503 https://doi.org/10.1016/j.envint.2020.105503.

Zaborowska, E., Czerwionka, K., Makinia, J., 2017. Strategies for achieving energy neutrality in biological nutrient removal systems – a case study of the Slupsk WWTP (northern Poland). Water Sci. Technol. 75, 727–740. https://doi.org/10.2166/wst.2016.564.

Zahedi, S., Gros, M., Balcazar, J.L., Petrovic, M., Pijuan, M., 2021. Assessing the occurrence of pharmaceuticals and antibiotic resistance genes during the anaerobic treatment of slaughterhouse wastewater at different temperatures. Sci. Total Environ. 789, 147910 https://doi.org/10.1016/j.scitotenv.2021.147910.

Zhang, J., Peng, Y., Li, X., Du, R., 2022. Feasibility of partial-denitrification/anammox for pharmaceutical wastewater treatment in a hybrid biofilm reactor. Water Res. 208, 117856 https://doi.org/10.1016/j.watres.2021.117856.

Zhang, L., Jiang, L., Zhang, J., Li, J., Peng, Y., 2022. Enhancing nitrogen removal through directly integrating anammox into mainstream wastewater treatment: Advantageous, issues and future study. Bioresour. Technol. 362, 127827 https://doi.org/10.1016/j.biortech.2022.127827.

Zhang, Q., Lin, J.-G., Kong, Z., Zhang, Y., 2022. A critical review of exogenous additives for improving the anammox process. Sci. Total Environ. 833, 155074 https://doi.org/10.1016/j.scitotenv.2022.155074.

Zhang, Z., Gao, P., Cheng, J., Liu, G., Zhang, X., Feng, Y., 2018. Enhancing anaerobic digestion and methane production of tetracycline wastewater in EGSB reactor with GAC/NZVI mediator. Water Res. 136, 54–63. https://doi.org/10.1016/j.watres.2018.02.025.

Zhang, Z., Yu, Y., Xi, H., Zhou, Y., 2021. Review of micro-aeration hydrolysis acidification for the pretreatment of toxic and refractory organic wastewater. J. Clean. Prod. 317, 128343 https://doi.org/10.1016/j.jclepro.2021.128343.

Zhao, W., Jeanne Huang, J., Hua, B., Huang, Z., Droste, R.L., Chen, L., Wang, B., Yang, C., Yang, S., 2020. A new strategy to recover from volatile fatty acid inhibition in anaerobic digestion by photosynthetic bacteria. Bioresour. Technol. 311, 123501 https://doi.org/10.1016/j.biortech.2020.123501.

Zhao, W., Yang, H., He, S., Zhao, Q., Wei, L., 2021. A review of biochar in anaerobic digestion to improve biogas production: Performances, mechanisms and economic assessments. Bioresour. Technol. 341, 125797 https://doi.org/10.1016/j.biortech.2021.125797.

Zhao, Y., Zhao, G., 2022. Decreasing ruminal methane production through enhancing the sulfate reduction pathway. Anim. Nutr. 9, 320–326. https://doi.org/10.1016/j.aninu.2022.01.006.

Zhen, G., Pan, Y., Lu, X., Li, Y.-Y., Zhang, Z., Niu, C., Kumar, G., Kobayashi, T., Zhao, Y., Xu, K., 2019. Anaerobic membrane bioreactor towards biowaste biorefinery and chemical energy harvest: recent progress, membrane fouling and future perspectives. Renew. Sustain. Energy Rev. 115, 109392 https://doi.org/10.1016/j.rser.2019.109392.

Zheng, Z., Ji, J., Hong, Y., Fang, Y., Wudi, Z., Xingling, Z., Changmei, W., Kai, W., Jing, L., 2022. Treatment of corn alcohol wastewater by anaerobic expanded granular sludge bed reactor and analysis of prokaryotic microbial community. Energy Sources Part Recovery Util. Environ. Eff. 44, 1830–1841. https://doi.org/10.1080/15567036.2019.1646842.

Zhou, Q., Jiang, X., Li, X., Jiang, W., 2016. The control of H2S in biogas using iron ores as in situ desulfurizers during anaerobic digestion process. Appl. Microbiol. Biotechnol. 100, 8179–8189. https://doi.org/10.1007/s00253-016-7612-7.

Zhou, Z., Ming, Q., An, Y., Ruan, D., Chen, G., Wei, H., Wang, M., Wu, Z., 2021. Performance and microbial community analysis of anaerobic sludge digestion enhanced by in-situ microaeration. J. Water Process Eng. 42, 102171 https://doi.org/10.1016/j.jwpe.2021.102171.

Zhuang, H., Zhu, H., Shan, S., Zhang, L., Fang, C., Shi, Y., 2018. Potential enhancement of direct interspecies electron transfer for anaerobic degradation of coal gasification wastewater using up-flow anaerobic sludge blanket (UASB) with nitrogen doped sewage sludge carbon assisted. Bioresour. Technol. 270, 230–235. https://doi.org/10.1016/j.biortech.2018.09.012.

Zhuang, H., Zhu, H., Zhang, J., Shan, S., Fang, C., Tang, H., Xie, Q., 2020. Enhanced 2,4,6-trichlorophenol anaerobic degradation by Fe3O4 supported on water hyacinth biochar for triggering direct interspecies electron transfer and its use in coal gasification wastewater treatment. Bioresour. Technol. 296, 122306 https://doi.org/10.1016/j.biortech.2019.122306.

Zuo, F., Yue, W., Gui, S., Sui, Q., Wei, Y., 2023. Resilience of anammox application from sidestream to mainstream: a combined system coupling denitrification, partial nitritation and partial denitrification with anammox. Bioresour. Technol. 374, 128783 https://doi.org/10.1016/j.biortech.2023.128783.

Żyłka, R., Karolinczak, B., Dąbrowski, W., 2021. Structure and indicators of electric energy consumption in dairy wastewater treatment plant. Sci. Total Environ. 782, 146599 https://doi.org/10.1016/j.scitotenv.2021.146599.